

22

Conjugate addition and nucleophilic aromatic substitution

Connections

➡ Building on

- Nucleophilic substitution at C=O **ch10** and at saturated C **ch15**
- Electrophilic additions to alkenes **ch19**
- Electrophilic substitution on aromatic rings **ch21**

Arriving at

- Conjugate addition: conjugation of alkenes with electron-withdrawing groups makes them electrophilic and allows nucleophilic attack
- Conjugate substitution: electrophilic alkenes bearing leaving groups can promote substitution reactions at C=C related to those at C=O
- Nucleophilic aromatic substitution: electron-poor aromatic rings that allow substitution reactions with nucleophiles rather than the usual electrophiles
- Special leaving groups and nucleophiles that allow nucleophilic aromatic substitution on electron-rich rings

➡ Looking forward to

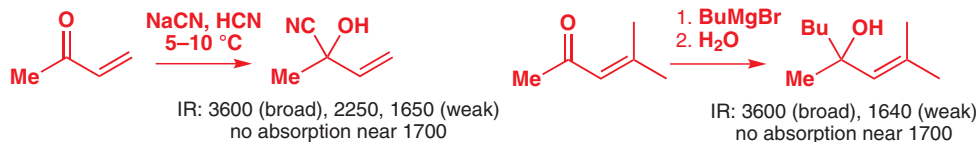
- Regioselectivity **ch24**
- Conjugate addition of enolates **ch26**
- Reactions of heterocyclic aromatic compounds **ch29 & ch30**

This chapter is also the last chapter in the second cycle of chapters within this book, with which we complete our survey of the important elementary types of organic reactions. We follow it with two review chapters, where we bring together aspects of *selectivity*, before looking in more detail at enolate chemistry and how to make molecules.

Alkenes conjugated with carbonyl groups

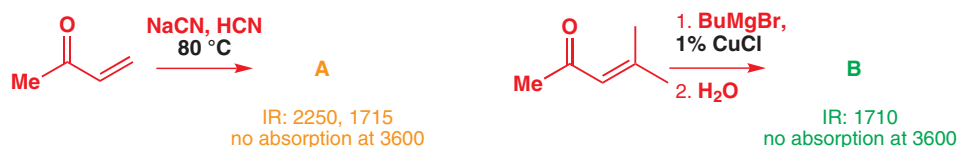
To start this chapter, let us take you back to one of the first reactions we introduced: nucleophilic addition to carbonyl groups. Here are two examples, both of which give products which you should fully expect. We've included details of the IR spectra of the products to confirm firstly that the product has no carbonyl group and secondly that the alkene is still there.

■ If you need to review IR spectroscopy, turn back to Chapter 3. Any C=O peak would appear near 1700 cm^{-1} , but there isn't one. Instead there's an O-H peak at 3600 cm^{-1} . The 2250 cm^{-1} peak is C≡N; C=C is at 1650 cm^{-1} .



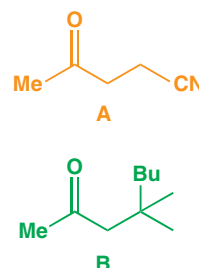
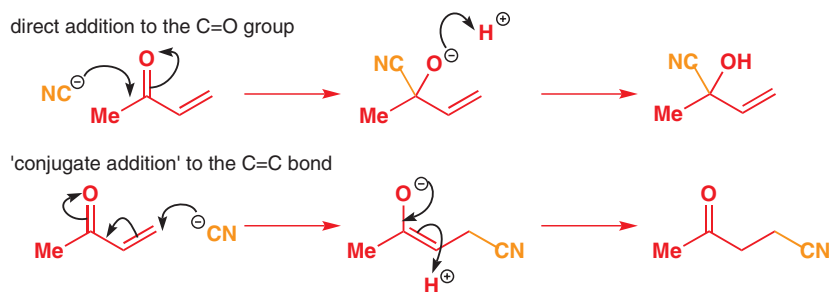
Now let's tweak the conditions: we repeat the first reaction at a higher temperature, and we add to the second a small amount of a copper salt. Now the products are different:

Online support. The icon in the margin indicates that accompanying interactive resources are provided online to help your understanding: just type www.chemtube3d.com/clayden/123 into your browser, replacing **123** with the number of the page where you see the icon. For pages linking to more than one resource, type **123-1**, **123-2** etc. (replacing **123** with the page number) for access to successive links.



Both products A and B have kept their carbonyl group (IR peak at 1710–1715 cm^{-1}) but have lost the C=C. Yet A, at least, is unquestionably an addition product because it contains a C $\equiv$ N peak at 2250 cm^{-1} .

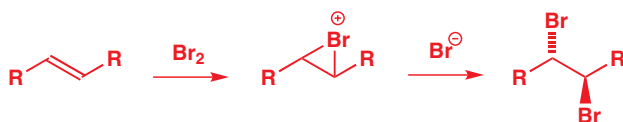
Well, the identities of A and B are revealed here: they are the products of addition, not to the carbonyl group, but to the C=C bond. Here's a mechanism, for both reactions of cyanide: firstly the direct addition to C=O and secondly addition to the C=C bond.



 Interactive mechanism for conjugate addition of cyanide

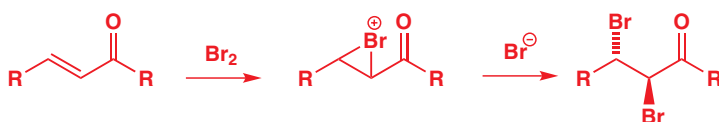
This type of reaction, where a nucleophile adds to a C=C double bond, is called **conjugate addition**, and this chapter is about the sorts of alkenes (and arenes) that do this kind of thing. We will also explain how such small differences in reaction conditions (temperature, or the presence of CuCl) manage to change the outcome so dramatically.

But first we need to place these conjugate additions into context. As you found out in Chapter 19, **alkenes are nucleophilic**. Almost regardless of their substituents, they react with electrophiles such as bromine to form adducts in which the π bond of the alkene has been replaced by two σ bonds.



 Reactions like this were discussed in Chapter 19.

Even when the alkene is conjugated with an electron-withdrawing group, as the alkenes on the last page were, bromine addition still occurs, although less readily. Never lose sight of this: alkenes are nucleophilic.



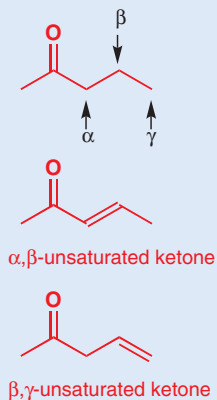
But as we have just seen, this last type of alkene also reacts with nucleophiles (such as cyanide, Grignard reagents, and, as you see below, more besides), and we now need to consider why.

Conjugated alkenes can be electrophilic

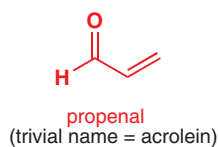
Conjugate additions occur only when the C=C double bond is immediately adjacent to the C=O group. They don't occur to C=C bonds that aren't conjugated (see the box on p. 501 for an illustration of this).

Compounds with double bonds adjacent to a C=O group are known as α,β -unsaturated carbonyl compounds. Many α,β -unsaturated carbonyl compounds have trivial names, and some are shown here. Some *classes* of α,β -unsaturated carbonyl compounds also have names such as 'enone', made up of 'ene' (for the double bond) + 'one' (for ketone).

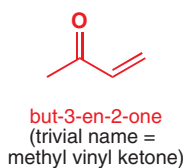
The α and β refer to the distance of the double bond from the $\text{C}=\text{O}$ group: the α carbon is the one next to $\text{C}=\text{O}$ (not the carbonyl carbon itself), the β carbon is one further down the chain, and so on.



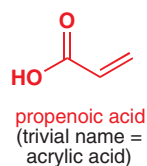
α,β -unsaturated aldehyde
(an enal)



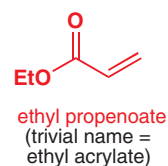
α,β -unsaturated ketone
(an enone)



α,β -unsaturated acid

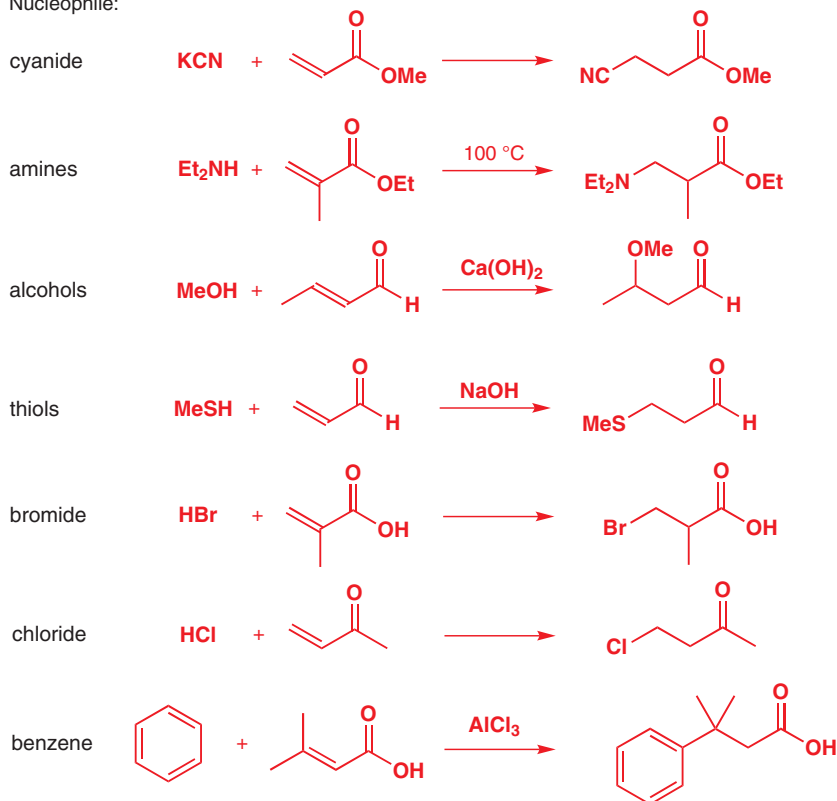


α,β -unsaturated ester



Most types of nucleophiles can be made to undergo conjugate additions with α,β -unsaturated carbonyl compounds, and seven examples are shown below. Note that many of these nucleophiles would not add to a simple carbonyl group: we will explain why shortly. Conjugate addition is also known as Michael addition, and the reactive α,β -unsaturated carbonyl compounds shown here are often known as Michael acceptors.

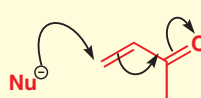
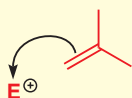
Nucleophile:



The reason that α,β -unsaturated carbonyl compounds react differently is conjugation, the phenomenon we discussed in Chapter 7. There we introduced you to the idea that bringing two π systems (two $\text{C}=\text{C}$ bonds, for example, or a $\text{C}=\text{C}$ bond and a $\text{C}=\text{O}$ bond) close together leads to a stabilizing interaction. It also leads to modified reactivity, because the π bonds no longer react as independent functional groups but as a single, conjugated system.

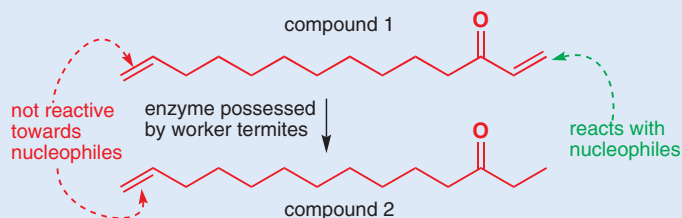
• Conjugation makes alkenes electrophilic

- $\text{C}=\text{C}$ double bonds are nucleophilic
- $\text{C}=\text{C}$ double bonds conjugated with carbonyl groups can be electrophilic



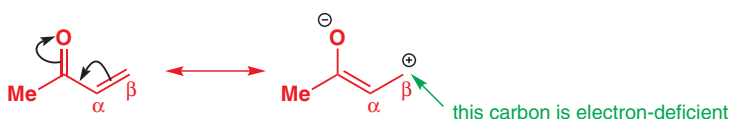
Termite self-defence and the reactivity of alkenes

Soldier termites of the species *Schedorhinotermes lamanianus* defend their nests with secretions of the enone shown below (compound 1), which is very effective at taking part in conjugate addition reactions with thiols (RSH). This makes it highly toxic, since many important biochemicals carry SH groups (one is described on p. 508). The worker termites of the same species—who build the nests—need to be able to avoid being caught in the crossfire, so they are equipped with an enzyme that allows them to reduce compound 1 to compound 2. This still has a double bond, but the double bond is completely unreactive towards nucleophiles because it is not conjugated with a carbonyl group. The workers escape unharmed.



Alkenes conjugated with carbonyl groups become polarized

To show why alkenes conjugated with carbonyl groups behave differently from unconjugated alkenes, we use curly arrows to indicate delocalization of the π electrons over the four atoms in the conjugated system. Both representations are extremes, and the true structure lies somewhere in between, but the polarized structure indicates why the conjugated C=C bond is electrophilic and why the β carbon is attacked by nucleophiles.



Polarization is detectable spectroscopically

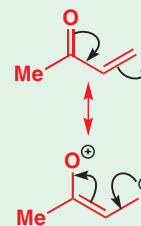
IR spectroscopy provides us with evidence for polarization in C=C bonds conjugated to C=O bonds. An unconjugated ketone C=O absorbs at 1715 cm^{-1} while an unconjugated alkene C=C absorbs (usually rather weakly) at about 1650 cm^{-1} . Bringing these two groups into conjugation in an α,β -unsaturated carbonyl compound leads to two peaks at 1675 and 1615 cm^{-1} , respectively, both quite intense. The lowering of the frequency of both peaks is consistent with a weakening of both π bonds (notice that the polarized structure has only single bonds where the C=O and C=C double bonds were). The increase in the *intensity* of the C=C absorption is consistent with polarization brought about by conjugation with C=O: a conjugated C=C bond has a significantly larger dipole moment than its unconjugated cousins.

The polarization of the C=C bond is also evident in the ^{13}C NMR spectrum, with the signal for the sp^2 carbon atom furthest from the carbonyl group moving downfield relative to an unconjugated alkene to about 140 ppm , and the signal for the other double bond carbon atom staying at about 120 ppm .

Molecular orbitals control conjugate additions

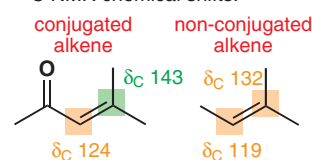
We have spectroscopic evidence that a conjugated C=C bond is polarized, and we can explain this with curly arrows, but the actual bond-forming step must involve movement of electrons from the HOMO of the nucleophile to the LUMO of the unsaturated carbonyl compound. This example is an efficient (the reaction happens even at 0°C) addition to acrolein (propenal) with methoxide as the nucleophile.

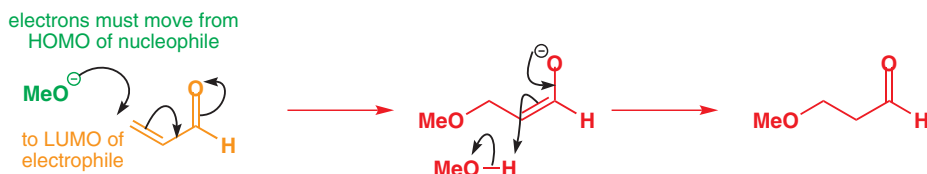
■ You may be asking yourself why we can't show the delocalization by moving the electrons the other way, like this.



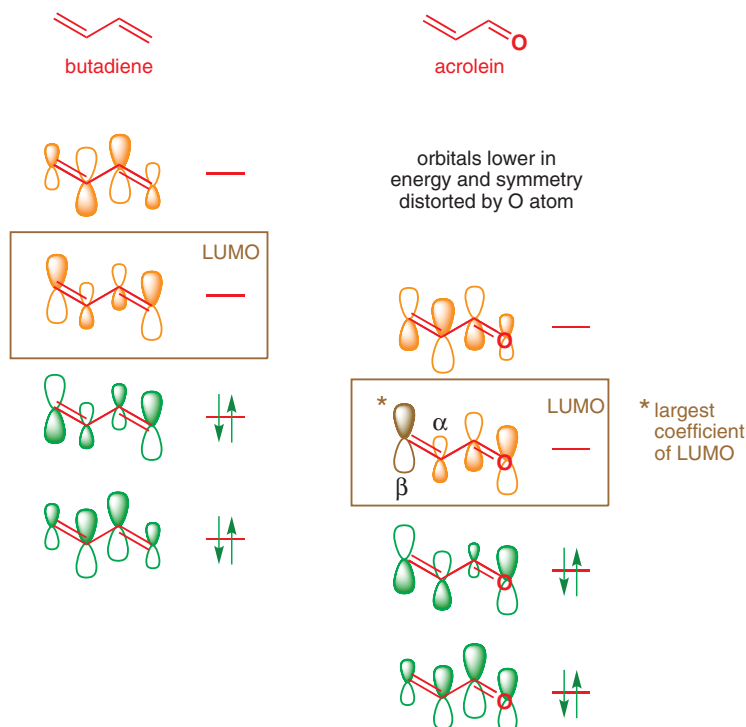
Think about electronegativities: O is much more electronegative than C, so it is quite happy to accept electrons, but here we have taken electrons away, leaving it with only six electrons. This structure therefore cannot represent the distribution of electrons in the conjugated system.

^{13}C NMR chemical shifts:





But what does this LUMO look like? It will certainly be more complicated than the π^* LUMO of a simple carbonyl group. The nearest thing you have met so far (in Chapter 7) are the orbitals of butadiene ($C=C$ conjugated with $C=C$), which we can compare with the α,β -unsaturated aldehyde acrolein ($C=C$ conjugated with $C=O$). The orbitals in the π systems of butadiene and acrolein are shown here. They are different because acrolein's orbitals are perturbed (distorted) by the oxygen atom (Chapter 4). You need not be concerned with exactly how the sizes of the orbitals are worked out, but for the moment just concentrate on the shape of the LUMO, the orbital that will accept electrons when a nucleophile attacks.



Interactive molecular orbitals in acrolein

In acrolein, the HOMO is in fact not the highest filled π orbital you see here, but the lone pairs on oxygen. This is not important, however, because here we are considering acrolein as an electrophile, so we are interested only in its LUMO.

In the LUMO, the largest coefficient is on the β carbon of the α,β -unsaturated system, shown with an asterisk. And it is here, therefore, that nucleophiles attack. In the reaction you have just seen, the HOMO is the methoxide oxygen's lone pair, so this will be the key orbital interaction that gives rise to the new bond.

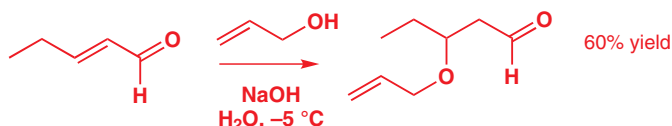


The second largest coefficient is on the $C=O$ carbon atom, so it's not surprising that some nucleophiles attack here as well—remember the example right at the beginning of the chapter where you saw cyanide attacking either the double bond or the carbonyl group depending on the conditions of the reaction. We shall next look at some conjugate additions with alcohols and amines as nucleophiles, before reconsidering the question of where the nucleophile attacks.

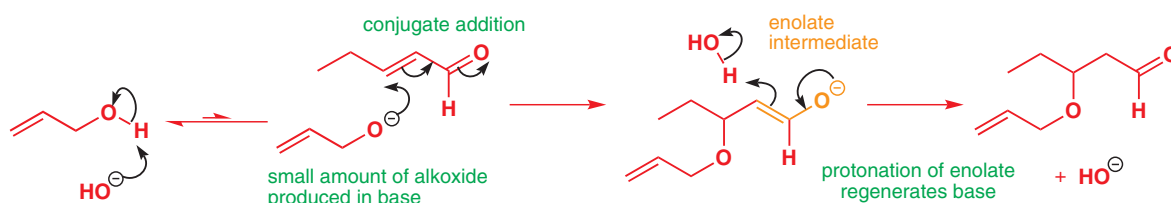
Conjugate additions have enolates or enols as intermediates

So much for the addition step of the reaction. But the product of this step is of course not the final product of the reaction—it is in fact an enolate. We hope you recognize these species from Chapter 20, where you saw them being made by treating carbonyl compounds with base. Conjugate addition is another way of generating an enolate, and as with all enolates, protonation gives back a carbonyl compound. The proton has to come from somewhere, so conjugate additions are usually done in protic solvents (such as alcohols or water). Here is another example with an alcohol as the nucleophile:

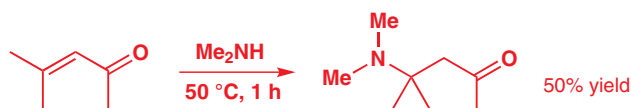
➡ Enols and enolates were introduced in Chapter 20.



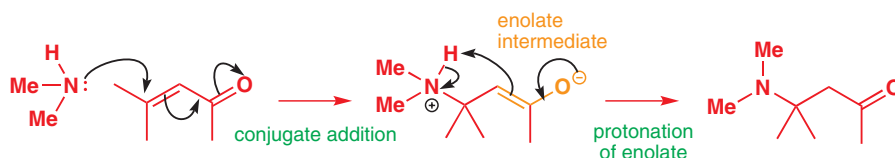
In alkaline solution, a small amount of alkoxide is produced (the pK_a of an alcohol is slightly higher than that of water), which attacks the C=C double bond in a conjugate addition. The product is an enolate, which is protonated by water to give the final aldehyde, and regenerates hydroxide as it does so: only a catalytic amount of base is required for this type of reaction.



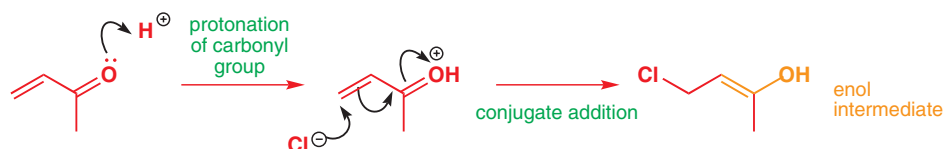
Amines are good nucleophiles for conjugate addition. In the reaction below, aqueous dimethylamine is used in a sealed system to stop the amine evaporating (dimethylamine is a gas even at room temperature).



Amines are neutral nucleophiles, and the amine itself provides a proton for the enolate.



If you survey the initial overview of conjugate additions on p. 500 you will see that several take place under acidic conditions. Treatment of this α,β -unsaturated ketone with HCl, for example, gives a chloroketone. The first step must be protonation of the carbonyl group, which makes the enone even more electrophilic by giving it a positive charge. Chloride attacks the β carbon to give an *enol*.



➡ Tautomerism is defined on p. 451.

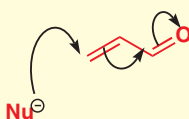
All that remains to happen now is tautomerism of the enol to its keto form by proton transfer from O to C.



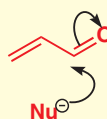
Conjugate addition or direct addition to the carbonyl group?

We have shown you several examples of conjugate additions using various nucleophiles and α,β -unsaturated carbonyl compounds, but we haven't yet addressed one important question. When do nucleophiles do conjugate addition (also called 1,4-addition) and when do they add directly to the carbonyl group (1,2-addition)? Several factors are involved—they are summarized here, and we will spend the next section of this chapter discussing them in turn.

- **Conjugate addition to C=C**
(also called 1,4-addition)



- **Direct addition to C=O**
(also called 1,2-addition)

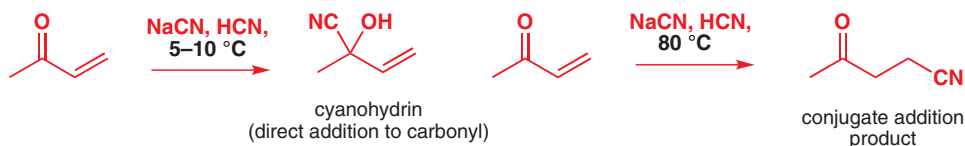


The way that nucleophiles react depends on:

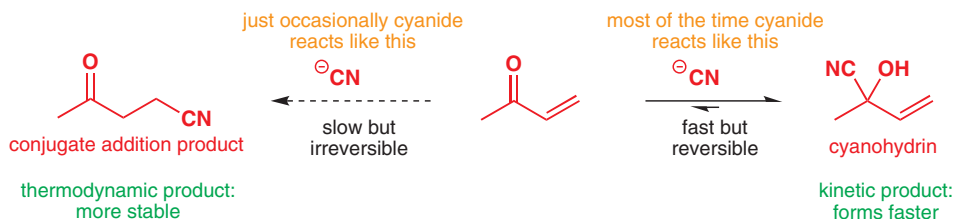
- the conditions of the reaction
- the nature of the α,β -unsaturated carbonyl compound
- the type of nucleophile.

Reaction conditions

The very first conjugate addition reaction in this chapter depended on the conditions of the reaction. Treating an enone with cyanide and an acid catalyst at low temperature gives a cyanohydrin by direct attack at C=O, while heating the reaction mixture leads to conjugate addition. What is going on?



We'll consider the low-temperature reaction first. As you know from Chapter 6, it is quite normal for cyanide to react with a ketone under these conditions to form a cyanohydrin. You also know from Chapter 6 that cyanohydrin formation is reversible. Even if the equilibrium for cyanohydrin formation lies well over to the side of the products, there will always be a small amount of starting enone remaining. Most of the time, this enone will react to form more cyanohydrin and, as it does, some cyanohydrin will decompose back to enone plus cyanide—such is the nature of a dynamic equilibrium. But every now and then—at a much slower rate—the starting enone will undergo a *conjugate addition* with the cyanide.



Now we have a different situation: conjugate addition is essentially an *irreversible* reaction, so once a molecule of enone has been converted to conjugate addition product, its fate is sealed: it cannot go back to enone again. Very slowly, therefore, the amount of conjugate addition product in the mixture will build up. In order for the enone–cyanohydrin equilibrium to be maintained, any enone that is converted to conjugate addition product will have to be replaced by reversion of cyanohydrin to enone plus cyanide. Even at room temperature, we can therefore expect the cyanohydrin to be converted bit by bit to conjugate addition product. This may take a very long time, but reaction rates are faster at higher temperatures, so at 80 °C this process does not take long at all and, after a few hours, the cyanohydrin has all been converted to conjugate addition product.

The contrast between the two products is this: the cyanohydrin is **formed faster** than the conjugate addition product, and is known as the product of kinetic control (or the kinetic product), but the conjugate addition product is the **more stable compound** and is the product of thermodynamic control (or the thermodynamic product). Typically, kinetic control involves lower temperatures and shorter reaction times, which ensures that only the fastest reaction has the chance to occur. And, typically, thermodynamic control involves higher temperatures and long reaction times to ensure that even the slower reactions have a chance to occur, and all the material is converted to the more stable compound.

➔ Kinetic and thermodynamic control were introduced in Chapter 12.

● Kinetic and thermodynamic control

- The product that forms faster is called the **kinetic product**.
- The product that is the more stable is called the **thermodynamic product**.

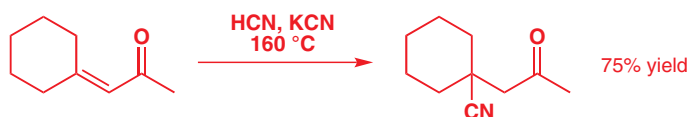
Similarly,

- Conditions that give rise to the kinetic product are called **kinetic control**.
- Conditions that give rise to the thermodynamic product are called **thermodynamic control**.

Why is direct addition faster than conjugate addition? Well, although the carbon atom β to the C=O group carries some positive charge, the carbon atom of the carbonyl group carries more, and so electrostatic attraction for the charged nucleophiles will encourage it to attack the carbonyl group directly rather than undergo conjugate addition.

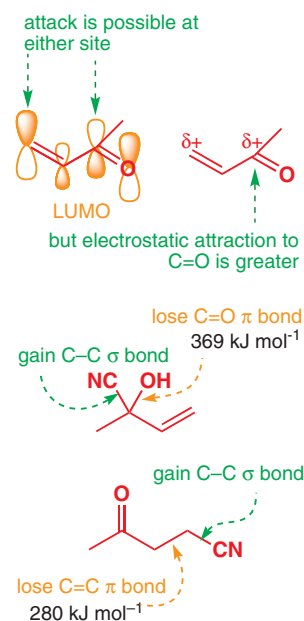
And why is the conjugate addition product the more stable? In the conjugate addition product, we gain a C–C σ bond, losing a C=C π bond, but keeping the C=O π bond. With direct addition, we still gain a C–C bond, but we lose the C=O π bond and keep the C=C π bond. C=O π bonds are stronger than C=C π bonds, so the conjugate addition product is more stable.

Practically, then, to get conjugate addition to occur you just have to give the reaction plenty of energy and maybe plenty of time to find its way to the most stable product. Here's an example: note the temperature!

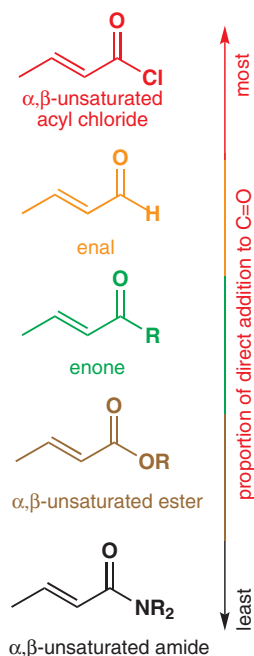
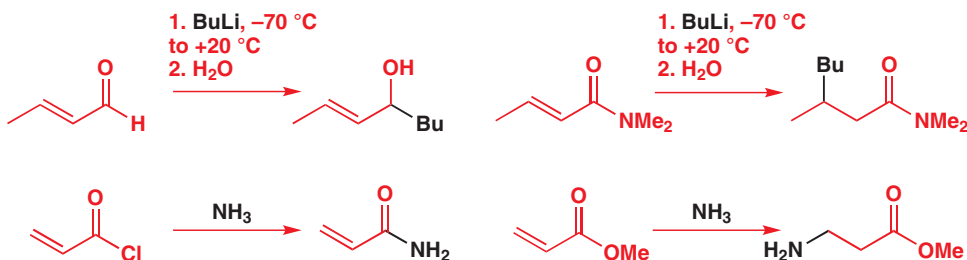


Structural factors

So far we have shown you conjugate additions mainly of α,β -unsaturated aldehydes and unsaturated α,β -ketones. You won't be at all surprised to learn, however, that unsaturated acids, esters, amides, and nitriles—in fact all carboxylic acid derivatives—can also take part in conjugate addition reactions. Two examples, an amide and an ester, are shown on the right below. But notice how the selectivity of these reactions depends on the structure of the unsaturated compound: compare the way butyllithium adds to this α,β -unsaturated aldehyde and α,β -unsaturated amide. Both additions are irreversible, and BuLi attacks the reactive carbonyl group of the aldehyde, but prefers conjugate addition to the less reactive amide. Similarly, ammonia reacts with this acyl chloride to give an amide product that derives from direct



addition to the carbonyl group, while with the ester it undergoes conjugate addition to give an amine.

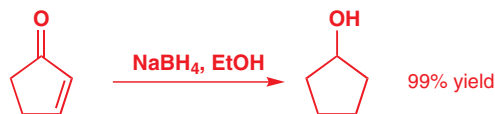


In both of these cases, the site of nucleophilic attack is determined simply by reactivity: the more reactive the carbonyl group, the more direct addition to C=O will result. The most reactive carbonyl groups, as you saw in Chapter 10, are those that are not conjugated with O or N (as they are in esters and amides), and particularly reactive are acyl chlorides and aldehydes. In general, the proportion of direct addition to the carbonyl group follows the reactivity sequence in the margin.

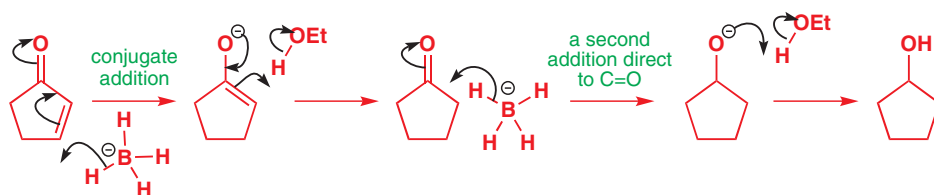
Sodium borohydride is a nucleophile that you have seen reducing simple aldehydes and ketones to alcohols, but it will also do conjugate addition reactions. Which of the alternatives actually takes place depends on the reactivity of the C=O group. NaBH₄ usually reacts with α,β-unsaturated aldehydes to give alcohols by direct addition to the carbonyl group.



Quite common with ketones, however, is the outcome below.



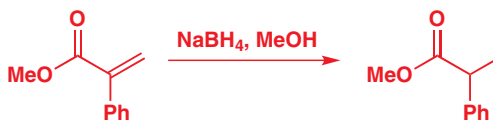
The borohydride has reduced not only the carbonyl group but the double bond as well. In fact, it's the double bond that's reduced first in a conjugate addition, followed by addition to the carbonyl group.



Luche reduction

It is possible to force NaBH₄ to attack only the C=O group by adding CeCl₃ to the reaction mixture. This modification is known as the Luche reduction, after its discoverer.

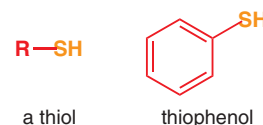
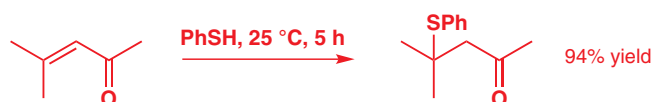
For esters and other less reactive carbonyl compounds conjugate addition is the only reaction that occurs because NaBH₄ doesn't reduce esters or amides.



The nature of the nucleophile: hard or soft

Among the best nucleophiles of all at doing conjugate addition are thiols, the sulfur analogues of alcohols. In this example, the nucleophile is thiophenol (phenol with the O replaced by S).

Remarkably, no acid or base catalyst is needed (as it was with the alcohol additions), and the product is obtained in 94% yield under quite mild reaction conditions.



■ We introduced the terms **hard** and **soft** in relation to nucleophiles in Chapter 15, p. 357,

So what's so special about a thiol? As you've seen already, attraction between nucleophiles and electrophiles is governed by two related interactions—electrostatic attraction between positive and negative charges and orbital overlap between the HOMO of the nucleophile and the LUMO of the electrophile. Successful reactions usually result from a combination of both, but sometimes reactivity can be dominated by one or the other. The dominant factor, be it electrostatic or orbital control, depends on the nucleophile and electrophile involved. Nucleophiles containing small, electronegative atoms (such as O or Cl), which we call 'hard', tend to react under predominantly electrostatic control, while 'soft' nucleophiles containing larger atoms (including the sulfur of thiols, but also P, I, and Se) are predominantly subject to control by orbital overlap.

The table below divides some nucleophiles into the two categories (plus some that lie in between)—but don't try to learn it! Rather, convince yourself that the properties of each one justify its location in the table. Most of these nucleophiles you have not yet seen in action, and the most important ones at this stage are indicated in **bold type**.

Hard and soft nucleophiles

Hard nucleophiles	Borderline	Soft nucleophiles
F^- , OH^- , RO^- , SO_4^{2-} , Cl^-	N_3^- , CN^-	I^- , RS^- , RSe^- , S^{2-}
H_2O , ROH , ROR' , $RCOR'$	RNH_2 , $R'R''NH$	RSH , RSR' , R_3P
NH_3 , $RMgBr$, RLi	Br^-	alkenes, aromatic rings

Not only can nucleophiles be classified as hard or soft, but electrophiles can too. For example, H^+ is a very hard electrophile because it is small and charged, while Br_2 is a soft electrophile: its orbitals are diffuse and it is uncharged. You saw Br_2 reacting with an alkene earlier in the chapter, and we explained in Chapter 5 that this reaction happens solely because of orbital interactions: no charges are involved.

● Hard/soft reactivity

- Reactions of hard species are dominated by charges and electrostatic effects.
- Reactions of soft species are dominated by orbital effects.
- Hard nucleophiles tend to react well with hard electrophiles.
- Soft nucleophiles tend to react well with soft electrophiles.

What has all this to do with the conjugate addition of thiols? Well, an α,β -unsaturated carbonyl compound is unusual in that it has two electrophilic sites, one of which is hard and one of which is soft. The carbonyl group has a high partial charge on the carbonyl carbon and will tend to react with hard nucleophiles, such as organolithium and Grignard reagents, that have a high partial charge on the nucleophilic carbon atom. Conversely, the β carbon of the α,β -unsaturated carbonyl system does not have a high partial positive charge but is the site of the largest coefficient in the LUMO. This makes the β carbon a soft electrophile and likely to react well with soft nucleophiles such as thiols.

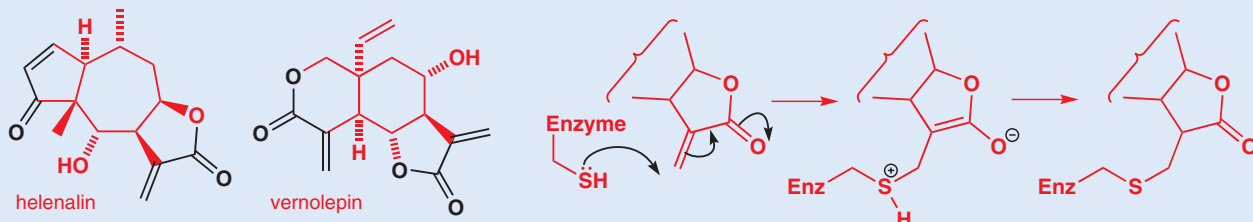
● Hard/soft—direct/conjugate addition

- Hard nucleophiles tend to react at the carbonyl carbon (hard) of an enone.
- Soft nucleophiles tend to react at the β carbon (soft) of an enone and lead to conjugate addition.

Anticancer drugs that work by conjugate addition of thiols

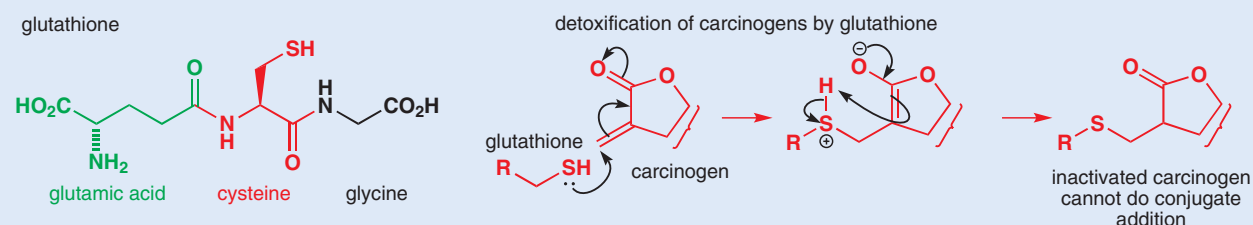
Drugs to combat cancer act on a range of biochemical pathways, but most commonly on processes that cancerous cells need to use to proliferate rapidly. One class attacks DNA polymerase, an enzyme needed to make the copy of DNA that has to be provided for each new cell. Helenalin and vernolepin are two such compounds, and if you look closely at their structure you should be able to spot

two α,β -unsaturated carbonyl groups in each. Biochemistry is just chemistry in very small flasks called cells, and the reaction between DNA polymerase and these drugs is simply a conjugate addition reaction between a thiol (the SH group of one of the enzyme's cysteine residues) and the unsaturated carbonyl groups. The reaction is irreversible and shuts down completely the function of the enzyme.



For this reason any compound capable of conjugate addition is potentially dangerous to living things. Even simple compounds like ethyl acrylate are labelled 'cancer suspect agents'. They attack enzymes, particularly the DNA polymerase involved in cell division by conjugate addition to thiol and amino groups in the enzyme. Fortunately, we are offered some degree of protection by an

important compound present in most tissues. The compound is glutathione, a tripeptide—a compound made from three amino acids. We shall discuss such compounds in more detail later in the book (Chapter 42) but notice for the moment that this compound can be divided into three at the two amide bonds.

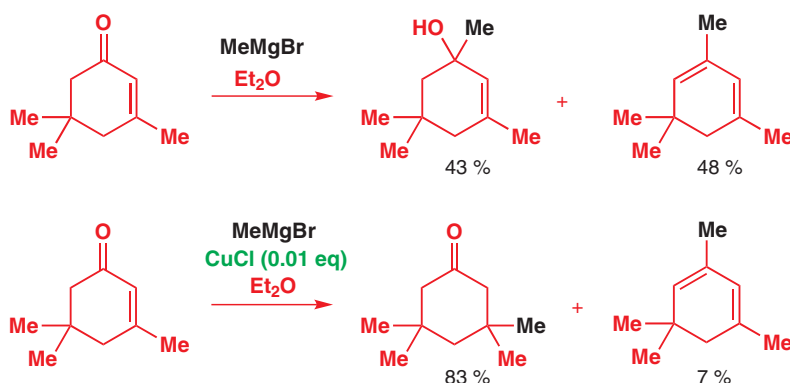


The business end of glutathione is the thiol (SH) group, which scavenges carcinogenic compounds by conjugate addition. If we use an 'exomethylene lactone'—a highly reactive Michael acceptor—as an example and represent glutathione as RCH_2SH , you can see the sort of thing that happens. If the normally abundant glutathione is removed by such processes as oxidation

(Chapter 42) and cannot any longer scavenge toxins, then the organism is in danger. This is one reason why 'antioxidants' like vitamin C are so beneficial—they remove stray oxidizing agents and protect the supply of glutathione. Keep eating the fruit and vegetables!

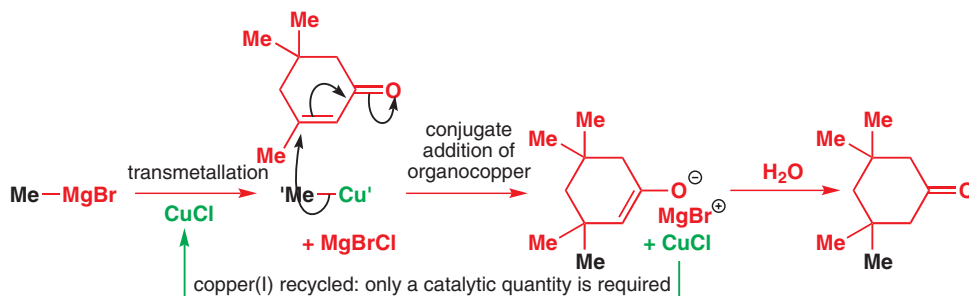
Promoting conjugate addition with copper(I) salts

Grignard reagents add directly to the carbonyl group of α,β -unsaturated aldehydes and ketones to give allylic alcohols: you have seen several examples of this, and you can now explain it by saying that the hard Grignard reagent prefers to attack the harder $C=O$ rather than the softer $C=C$ electrophilic centre. Here is a further example—the addition of $MeMgBr$ to a cyclic ketone to give an allylic alcohol, plus, as it happens, some of a diene that arises from this alcohol by loss of water (dehydration). Below this example is the same reaction to which a very small amount (just 0.01 equivalents, that is, 1%) of copper(I) chloride has been added. The effect of the copper is dramatic: it makes the Grignard reagent undergo conjugate addition, with only a trace of the diene.



Organocopper reagents undergo conjugate addition

The copper works by *transmetallating* the Grignard reagent to give an organocopper reagent—simply put, the magnesium is exchanged for copper. Organocoppers are softer than Grignard reagents, and add in a conjugate fashion to the softer C=C double bond. Once the organocopper has added, the copper salt is available to transmetallate some more Grignard, and only a catalytic amount is required.

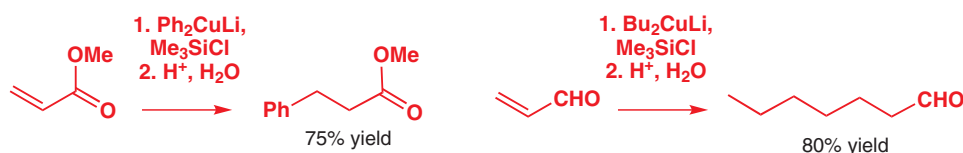


■ Organocoppers are softer than Grignard reagents because copper is less electropositive than magnesium, so the C–Cu bond is less polarized than the C–Mg bond, giving the carbon atom less of a partial negative charge. Electronegativities: Mg, 1.3; Cu, 1.9.

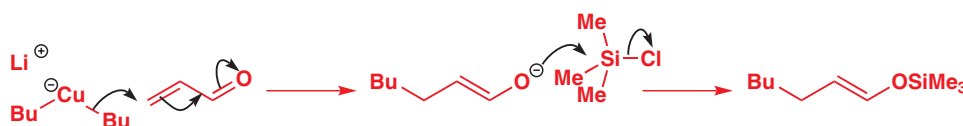
The organocopper is shown here as ‘Me–Cu’ because its precise structure is not known. But there are other organocopper reagents that also undergo conjugate addition and are much better understood. The simplest result from the reaction of two equivalents of organolithium with one equivalent of a copper (I) salt such as CuBr in ether or THF solvent at low temperature. The lithium cuprates (R_2CuLi) that are formed are not stable and must be used immediately.



The addition of lithium cuprates to α,β -unsaturated ketones turns out to be much better if trimethylsilyl chloride is added to the reaction—we shall explain what this does shortly, but for the moment here are two examples of lithium cuprate additions.



The silicon works by reacting with the negatively charged intermediate in the conjugate addition reaction to give a silyl enol ether—a type of molecule we met in Chapter 20. Here is a possible mechanism for a reaction between Bu_2CuLi and an α,β -unsaturated aldehyde in the presence of Me_3SiCl . The silyl enol ether simply hydrolyses to the ketone at the end of the reaction.



Summary: factors controlling conjugate addition

At this point in the chapter it is worthwhile talking stock of the factors controlling the two modes of addition to α,β -unsaturated carbonyl compounds.

● Conjugate (1,4 or Michael) vs direct (1,2) addition

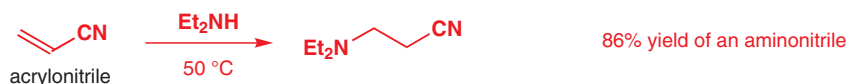
	Conjugate addition favoured by	Direct addition to C=O favoured by
Reaction conditions (for reversible additions):	thermodynamic control: high temperatures, long reaction times	kinetic control: low temperatures, short reaction times
Structure of α,β -unsaturated compound:	unreactive C=O group (amide, ester) unhindered β carbon	reactive C=O group (aldehyde, acyl chloride) hindered β carbon
Type of nucleophile:	soft nucleophiles	hard nucleophiles
Organometallic:	organocoppers or catalytic Cu(I)	organolithiums, Grignard reagents

Extending the reaction to other electron-deficient alkenes

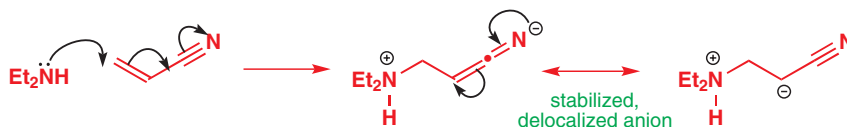
It's not only carbonyl-based groups that make alkenes react with nucleophiles rather than the more usual electrophiles. Other electron-withdrawing groups do just the same thing. Here are two examples: a nitrile and a nitro group. These compound classes appeared in Chapter 21 in the context of aromatic substitution reactions where we saw them pulling electron density away from the ring. The same thing happens here.

Unsaturated nitriles and nitro compounds

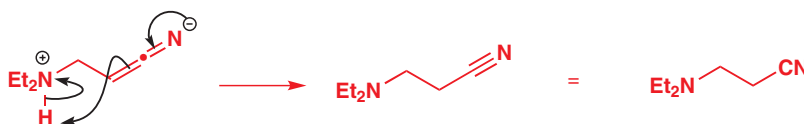
The simplest conjugated nitrile is acrylonitrile. This compound adds amines readily. No special conditions are needed to encourage attack at C=C rather than C $\equiv$ N because the nitrile carbon is rather unreactive as an electrophilic centre.



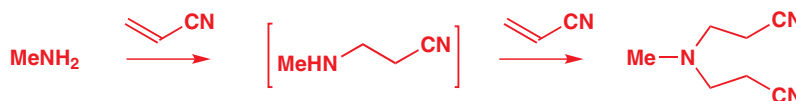
The amine first attacks the alkene in a typical conjugate addition to make an anion stabilized by being next to the nitrile. The anion can have its charge drawn on C or N: it is delocalized like an enolate. Do not be put off by the odd appearance of the 'enolate'. The dot between the two double bonds is a reminder that there is a linear sp carbon atom at this point.



Protonation at carbon restores the nitrile and gives the product—an amino-nitrile. The whole process adds a 2-cyano-ethyl group to the amine and is known industrially as cyanoethylation.



With a primary amine, the reaction need not stop at that stage as the product is still nucleophilic and a second addition can occur to replace the second hydrogen atom on nitrogen.



➡ For reactions where strong nucleophiles do attack C $\equiv$ N—partly because they have nothing else to attack—see pp. 220 and 231.

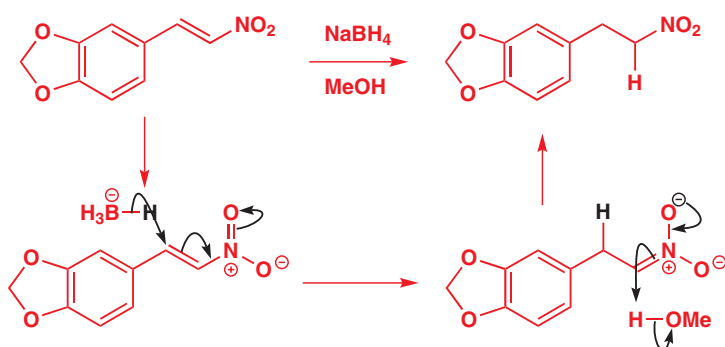
🖱 Conjugate addition of amines to acrylonitrile

■ You will see a few mechanisms in this chapter where we have written an intramolecular deprotonation. This saves writing two steps—protonation of the enolate and deprotonation of N (in this case)—but quite possibly this is not the actual mechanism by which the proton transfer takes place. Any proton will do, as will any base—the protons are hopping around all the time, so as with any proton transfer it doesn't pay to take the arrows too literally. We discussed alternative but equivalent mechanisms for proton transfers in Chapter 12, p. 267.

Other elements such as O, S, or P can add too. Phenyl phosphine can undergo a double addition just as in the last example, but alcohols can add only once. If there is competition between a first-row (for example N or O) and a second-row (for example S or P) element, the second-row element normally wins, for the reasons discussed above (p. 507).

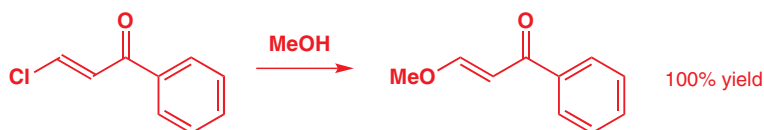


The nitro group (NO₂) is extremely electron-withdrawing—about twice as electron-withdrawing as a carbonyl group. It is also unreactive as an electrophilic centre, which makes conjugate addition to nitro-alkenes a very reliable reaction. In this example, sodium borohydride attacks the C=C bond in a conjugate manner to give an intermediate looking rather like an enolate anion, with a negatively charged oxygen atom conjugated to an (N=C) double bond. It reacts like an enolate too, picking up a proton on carbon to re-form the nitro group and give a stable product.

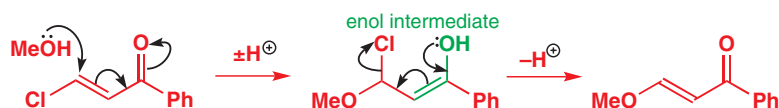


Conjugate substitution reactions

Just as direct addition to C=O (Chapter 6) becomes substitution at C=O (Chapter 10) when there is a leaving group at the carbonyl carbon, so conjugate *addition* becomes conjugate *substitution* if there is a leaving group, such as Cl, at the β carbon atom. Here is an example: substitution replaces Cl with OMe, just as it would have done in a reaction with an acyl chloride.



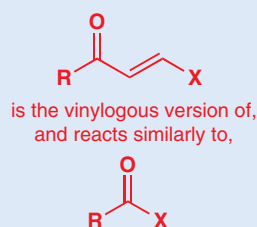
As with substitution at C=O, this apparently simple reaction does *not* involve a direct displacement of the leaving group in a single step. The mechanism starts in exactly the same way as for conjugate addition, giving an enol intermediate.



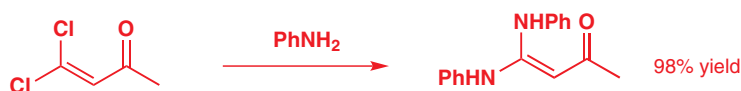
Now the leaving group can be expelled by the enol: the double bond moves back into its original position in an elimination reaction—the sequence is often called an addition–elimination

Vinylogous behaviour

Compounds like this are known as vinylogous amides—the conjugated double bond serves as an electronic linker between the carbonyl group and the heteroatom, which makes the chemical and spectroscopic behaviour of the composite functional group similar to that of its simpler relative. You could think of the β -chloro enone at the beginning of this section as a vinylogous acyl chloride that reacts with methanol to give a vinylogous ester.



reaction. The 'new' double bond has the more stable *E* configuration. In the next example, two consecutive conjugate substitution reactions give a 1,1-diamine.



At first sight, the product looks rather unstable—sensitive to water, or traces of acid perhaps. But, in fact, it is remarkably resistant to reaction with both. The reason is conjugation: this isn't really an amine (or a diamine) at all because the lone pairs of the nitrogen atoms are delocalized through into the carbonyl group, very much as they are in an amide. This makes them less basic, and makes the carbonyl group less electrophilic.

delocalization of the nitrogen's lone pair

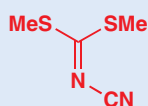
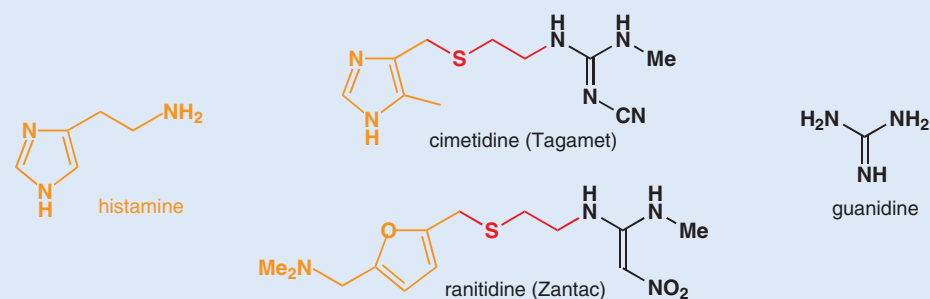


delocalization in an amide

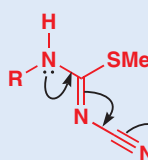
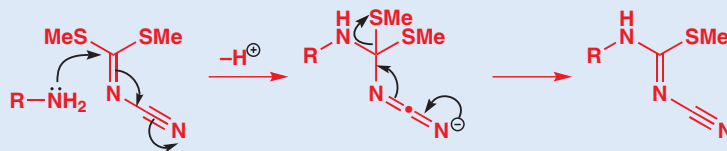


Conjugate substitution and the synthesis of anti-ulcer drugs

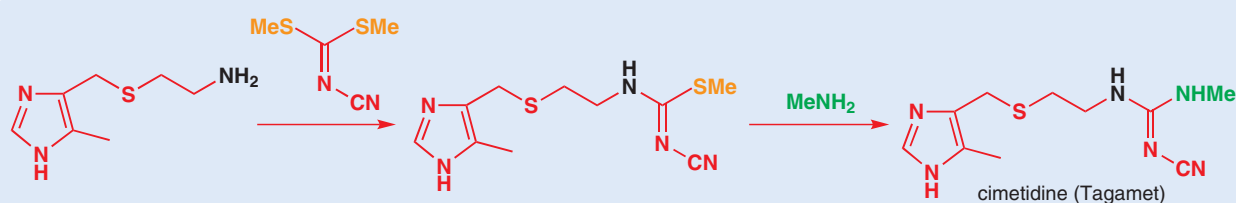
Just as the cyanide (CN) and nitro (NO₂) groups can be used to bring about conjugate addition, so also can they initiate conjugate substitution. Examples of these reactions play important roles in the synthesis of two of the most significant drugs in the development of modern medicinal chemistry: the anti-ulcer compounds cimetidine (marketed as Tagamet) and ranitidine (Zantac). We looked at some aspects of the structure of these drugs in Chapter 8 (p. 178) and we are now going to see how conjugate addition is used in their synthesis.



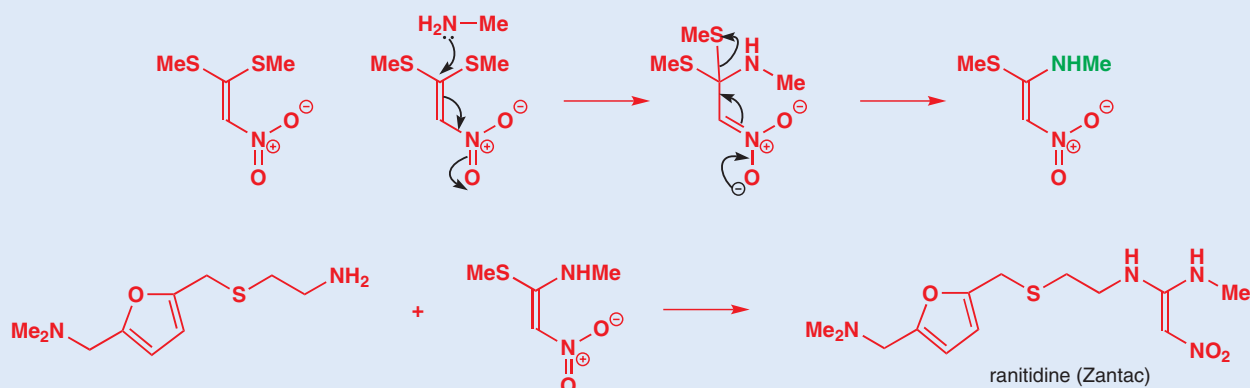
The simple cyanoimine on the left, with two SMe groups as built-in leaving groups, is readily available and reacts with amines to give guanidines in two stages. Each of the reactions is a conjugate substitution. It will be clearer if we draw the reaction with a generalized primary amine RNH₂ first: conjugate addition, exactly as we saw with acrylonitrile, is followed by expulsion of the best leaving group. Thiols are acidic compounds, and MeS[−] is a better leaving group than RNH[−].



The reaction stops cleanly at this point and more vigorous conditions are required to displace the second MeS[−] group. This is because the first product is less reactive than the starting material: the new amino group is electron-donating and conjugation is established between it and the cyano group, deactivating the molecule towards a second conjugate substitution (as shown in the margin). But under more forcing conditions a second and different amine can be introduced and the second MeS[−] group displaced. In the synthesis of cimetidine the second amine is MeNH₂ and the molecule is complete.



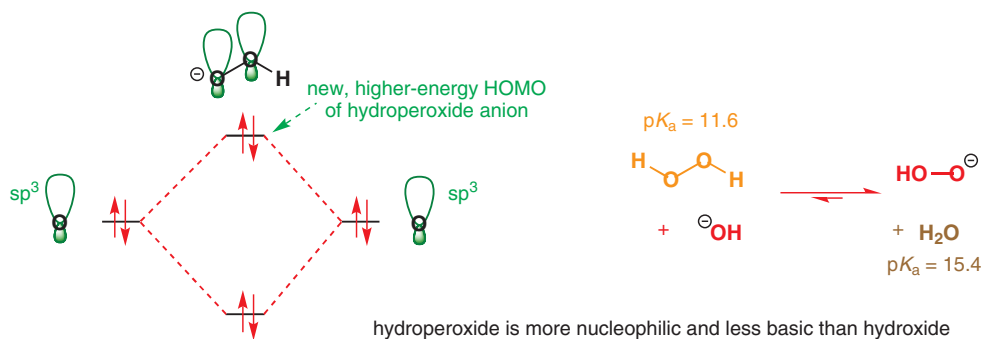
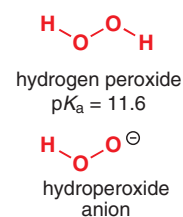
Ranitidine's right-hand portion is made in a similar way from an unsaturated nitro compound. This time the methylamine substitution is done first, followed by addition of the rest of the molecule.



Nucleophilic epoxidation

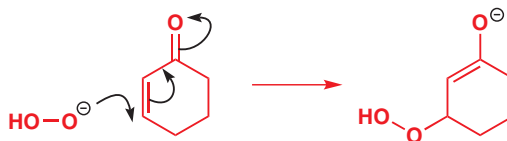
The conjugate substitutions we have just been discussing rely on a starting material containing a leaving group. In this section we are going to look at what happens if the leaving group is not attached to the unsaturated carbonyl compound, but instead is attached to the nucleophile. We shall look at this class of compounds—nucleophiles with leaving groups attached—in more detail in Chapter 38), but for the moment the most important will be hydroperoxide, the anion of hydrogen peroxide.

Hydroperoxide is a good nucleophile because of the **alpha effect**: interaction of the two lone pairs on adjacent oxygen atoms raises the HOMO of the anion and makes it a better and softer nucleophile than hydroxide. Hydroperoxide is also less basic than hydroxide because of the inductive electron-withdrawing effect of the second oxygen atom. Basicity and nucleophilicity usually go hand in hand—not here though. This means that the hydroperoxide anion can be formed by treating hydrogen peroxide with aqueous sodium hydroxide.

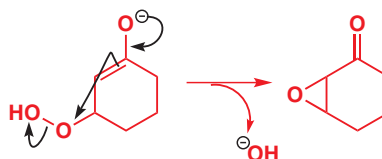


This is what happens when this mixture is added to an enone. First, there is the conjugate addition.

■ The same effect explains why hydroxylamine and hydrazine are more nucleophilic than ammonia (see p. 232).



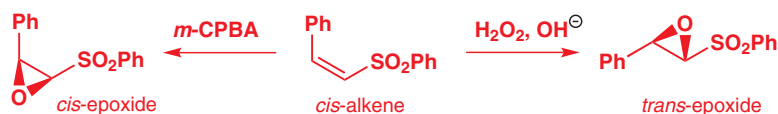
But the product is not stable because hydroxide can be lost from the oxygen atom that was the nucleophile. Hydroxide is fine as a leaving group here—after all, hydroxide is lost from enolates in E1cB eliminations, and here the bond breaking is a weak O—O bond. The product is an epoxide.



Interactive mechanism for nucleophilic epoxidation

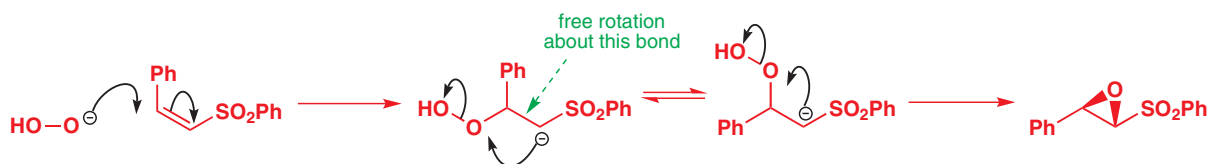
The electrophilic epoxidizing agents such as *m*-CPBA, which you met in Chapter 19, work reliably only with nucleophilic alkenes, and for α,β -unsaturated carbonyl compounds and other electron-deficient alkenes, hydroperoxide—a nucleophilic epoxidizing agent—is often used instead.

There is another significant difference between hydrogen peroxide and *m*-CPBA, highlighted by the pair of reactions below.



➡ For a reminder of the meaning of the term *stereospecific*, see p. 396.

m-CPBA epoxidation is stereospecific because the reaction happens in one step. But nucleophilic epoxidation with hydroperoxide is a two-step reaction: there is free rotation about the bond marked in the anionic intermediate, and the more stable, *trans*-epoxide results, whatever the geometry of the starting alkene.

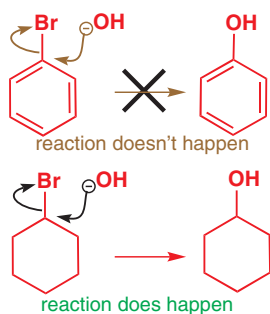


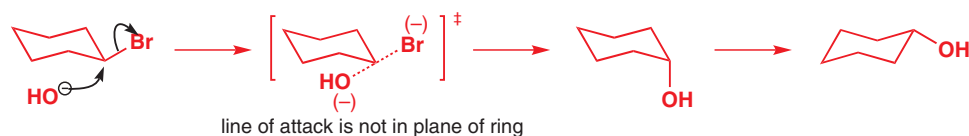
Nucleophilic aromatic substitution

In this next section we are going to consider reactions related to conjugate substitutions but in which the double bond is part of an aromatic ring. We spent some considerable time in Chapter 21 explaining that aromatic rings are *nucleophilic*: electrophiles attack them, and typical aromatic reactivity is to undergo electrophilic substitution.

In general, nucleophilic substitutions of aromatic halides—such as the one proposed here in which hydroxide is attempting to displace bromide—**do not happen**. You might well ask, ‘Why not?’ The reaction looks all right and, if the ring were saturated, it *would be* all right.

This is an S_N2 reaction, and we know (Chapter 15) that attack must occur in line with the C—Br bond from the back, where the largest lobe of the σ^* orbitals lies. That is perfectly all right for the aliphatic ring because the carbon atom is tetrahedral and the C—Br bond is not in the plane of the ring. Substitution of an equatorial bromine goes like this:





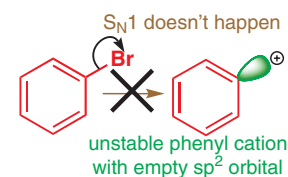
But in the aromatic compound, the C–Br bond *is* in the plane of the ring as the carbon atom is trigonal. To attack from the back, the nucleophile would have to appear inside the benzene ring and invert the carbon atom in an absurd way. This reaction is of course not possible.

This is another example of the general rule:



● S_N2 at sp^2 C does not occur.

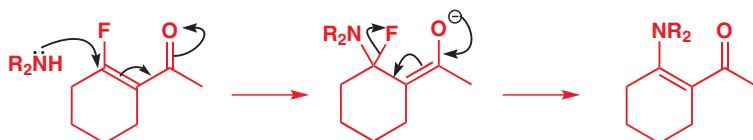
If S_N2 is impossible, what about S_N1 ? This is possible but very unfavourable unless the leaving group is an exceptionally good one (see below for an example). It would involve the unaided loss of the leaving group and the formation of an aryl cation. All the cations we saw as intermediates in the S_N1 reaction (Chapter 15) were planar with an empty p orbital. This cation is planar but the p orbital is full—it is part of the aromatic ring—and the empty orbital is an sp^2 orbital outside the ring.



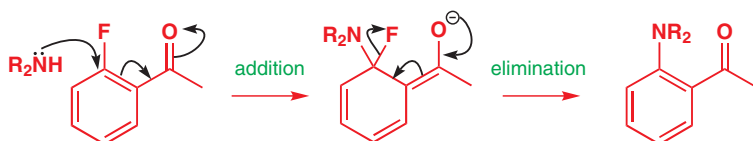
Yet some aromatic compounds *do* undergo nucleophilic substitution. Just as normally nucleophilic alkenes can be made to undergo conjugate substitution if they carry electron-withdrawing substituents, so normally nucleophilic aromatic rings also become electrophilic if they have the right substituents. The mechanism by which they undergo nucleophilic substitution also closely parallels that of conjugate substitution which you have just seen.

The addition–elimination mechanism

Imagine a cyclic β -fluoro-enone reacting with a secondary amine in a conjugate substitution reaction. The normal addition to form the enolate followed by return of the negative charge to expel the fluoride ion gives the product.



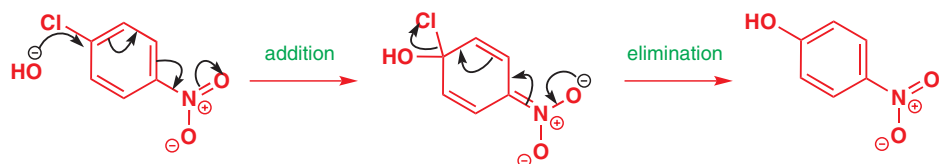
Now imagine just the same reaction with two extra double bonds in the ring. These play no part in our mechanism; they just make what was an aliphatic ring into an aromatic one. Conjugate substitution has become **nucleophilic aromatic substitution**.



The mechanism involves *addition* of the nucleophile followed by *elimination* of the leaving group—the **addition–elimination mechanism**. It is not necessary to have a carbonyl group—any electron-withdrawing group will do—the only requirement is that the electrons must be able to get out of the ring into this anion-stabilizing group. Here is an example with a *para*-nitro group.

■ This mechanism is also abbreviated to S_NAr (for Substitution, Nucleophilic, Aromatic).

Interactive mechanism for aromatic addition–elimination

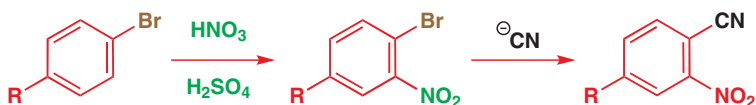


Everything is different about this example—the nucleophile (HO^-), the leaving group (Cl^-), the anion-stabilizing group (NO_2), and its position (*para*)—but the reaction still works. The nucleophile is a good one, the negative charge can be pushed through on to the oxygen atom(s) of the nitro group, and chloride is a better leaving group than OH^- .

• A typical nucleophilic aromatic substitution has:

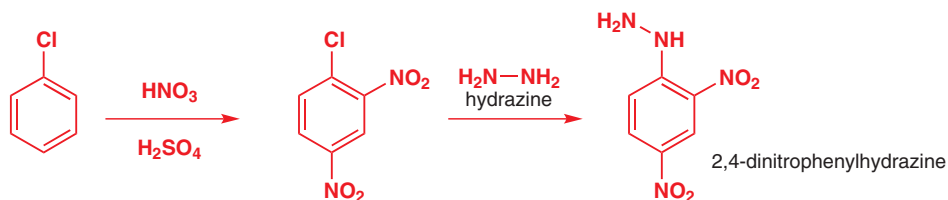
- an oxygen, nitrogen, or cyanide nucleophile
- a halide for a leaving group
- a carbonyl, nitro, or cyanide group *ortho* and/or *para* to the leaving group.

Since the *nitro* group is usually introduced by electrophilic aromatic substitution (Chapter 21) and halides direct *ortho/para* in nitration reactions, a common sequence is nitration followed by nucleophilic substitution.



This sequence is useful because the nitro group could not be added directly to give the final product as nitration would go in the wrong position. The nitrile is *meta*-directing, while the alkyl group (R) is *ortho, para*-directing.

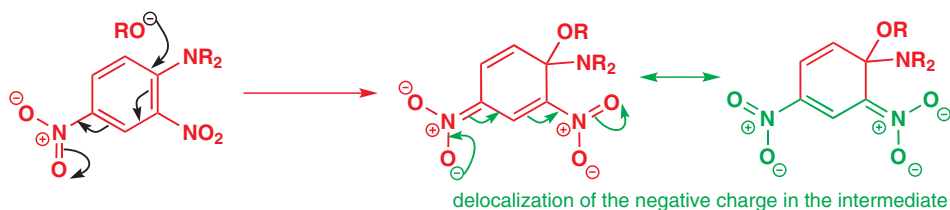
Two activating electron-withdrawing groups are better than one and dinitration of chlorobenzene makes a very electrophilic aryl halide. Reaction with hydrazine gives a useful reagent.



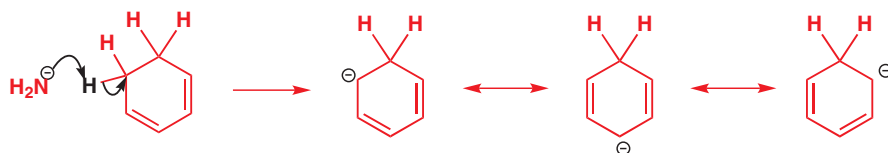
It also makes a very toxic one! This compound—2,4-dinitrophenylhydrazine—is carcinogenic. Nonetheless it forms coloured crystalline imines (hydrazones) with carbonyl compounds—before the days of spectroscopy these were used to characterize aldehydes and ketones (see p. 232).

The intermediate in the addition–elimination mechanism

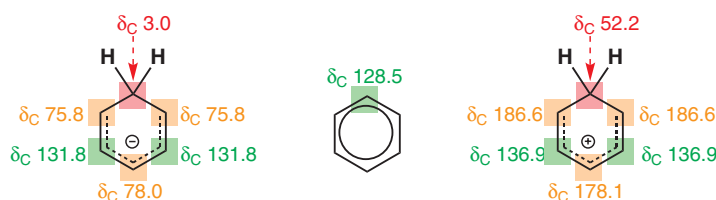
What evidence is there for intermediates like the ones we have been using in this section? When reactions like this last example are carried out, a purple colour often appears in the reaction mixture and then fades away. In some cases the colour is persistent and thought to be due to the intermediate. Here is an example with RO^- attacking a nitrated aniline. This intermediate is persistent because neither potential leaving group (NR_2 or OR) is very good.



What is the nature of this intermediate? Well, in essence it is an anion delocalized over five sp^2 hybridized carbons of a six-membered ring (the sixth, the point at which the nucleophile attacked, is sp^3 hybridized). It's possible to make a simple homologue of such a species by deprotonating cyclohexadiene. Delocalizing the anion generates the three structures below.



You've seen before that ^{13}C NMR spectra are revealing when it comes to distribution of charge, and the details of the ^{13}C NMR spectrum of this anion are shown below, along with those of benzene itself and also of the cation generated by protonating benzene (which, as you will remember from Chapter 21, corresponds to the intermediate generated in electrophilic aromatic substitution).

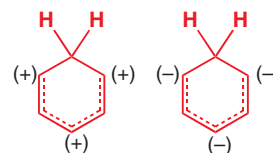
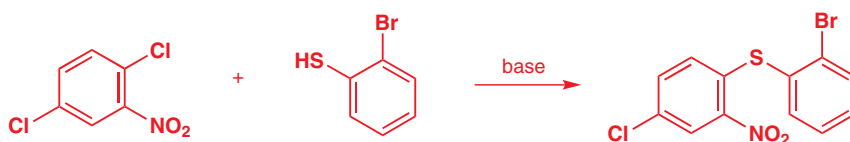


■ A reminder: A larger shift means less electronic shielding and a smaller shift more electronic shielding.

These results are very striking. The shifts of the *meta* carbons in both ions are very slightly different from those of benzene itself (about 130 ppm). But the *ortho* and *para* carbons in the anion have gone upfield to much smaller shifts, indicating greater electron density. By contrast, *ortho* and *para* carbons in the cation have gone downfield to much larger shifts.

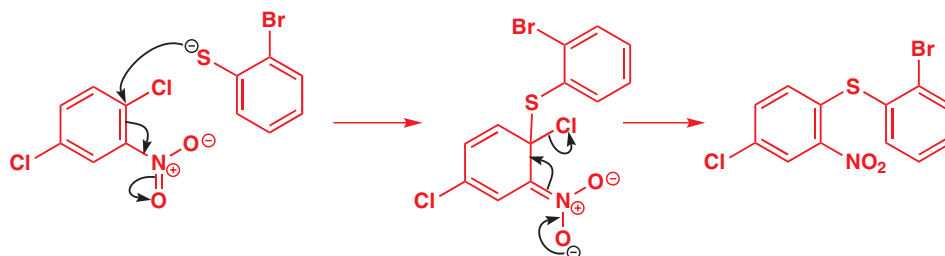
The differences are very great—about 100 ppm between the cation and the anion! It is very clear from these spectra that the ionic charge is delocalized almost exclusively to the *ortho* and *para* carbons in both cases. The alternative structures in the margin show this delocalization.

This means that stabilizing groups, such as nitro or carbonyl in the case of the anion, can only have an effect if they are on carbons *ortho* or *para* to the position being attacked by the nucleophile. A good illustration of this is the selective displacement of one chlorine atom out of these two. The chlorine *ortho* to the nitro group is lost; the one *meta* is retained.



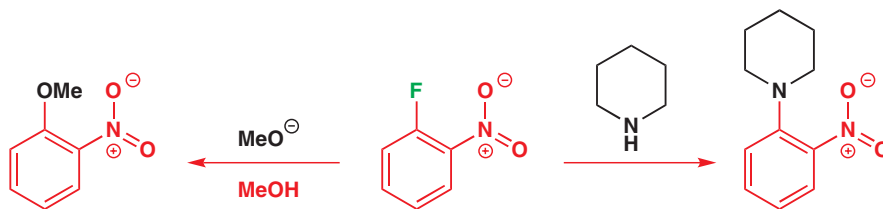
■ Remember, charges in brackets show significant (here ca. 1/3) portions of a charge, in contrast with δ , which means a much smaller polarization.

The mechanism works well if the nucleophile (the anion derived from the thiol) attacks the carbon bearing the chlorine *ortho* to the nitro group as the negative charge can then be pushed into the nitro group. Satisfy yourself that you cannot do this if you attack the other chlorine position. This is a very practical reaction and is used in the manufacture of a tranquillizing drug.

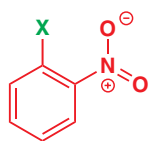


The leaving group and the mechanism

In the first nucleophilic aromatic substitution that we showed you, we used fluoride ion as a leaving group. Fluoride works very well in these reactions and even such a simple compound as 2-halo-1-nitrobenzenes reacts efficiently with a variety of nucleophiles, as in these examples.



reactivity of 2-halo-1-nitrobenzenes in nucleophilic aromatic substitution

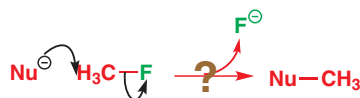


$F \gg Cl \sim Br \gg I$

The same reactions happen with the other 2-halo-1-nitrobenzenes but less efficiently. The fluoro-compound reacts about 10^2 – 10^3 times faster than the chloro or bromo compounds and the iodo compound is even slower.

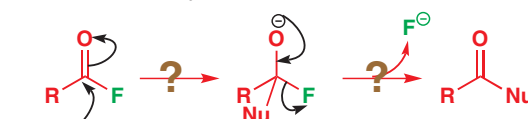
This ought to surprise you. When we were looking at other nucleophilic substitutions such as those at the carbonyl group or saturated carbon, we never used fluoride as a leaving group! The C–F bond is very strong—the strongest of all the single bonds to carbon—and it is difficult to break. As a consequence, these reactions are not a good prospect:

This reaction is never used:



use instead Cl, Br, or I (I is best)

This reaction is rarely used:



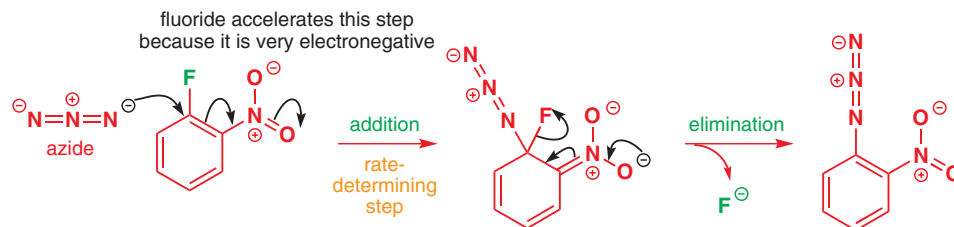
use instead Cl

So why is fluoride so good in nucleophilic aromatic substitution when the reverse is true with other reactions? You will notice that we have *not* said that fluoride is a better leaving group in nucleophilic aromatic substitution. It isn't! The explanation depends on a better understanding of the mechanism of the reaction. We shall use azide ion as our nucleophile because this has been well studied and because it is one of the best.

The mechanism is exactly the same as that we have been discussing all along—a two-stage addition–elimination sequence. In a two-step mechanism, one step is slower and rate determining; the other is unimportant to the rate. You may guess that, in the mechanism for nucleophilic aromatic substitution, it is the first step that is slower because it disturbs the aromaticity. The second step restores the aromaticity and is faster. The effect of fluoride, or any other leaving group, can only come from its effect on the first step. How good a leaving group it might be does not matter: the rate of the second step—the step where fluoride leaves—has no effect on the overall rate of the reaction.

➡ You met the azide anion in Chapter 15, p. 354.

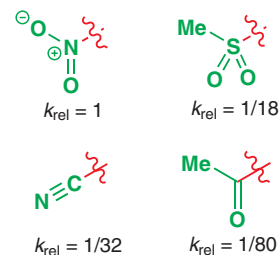
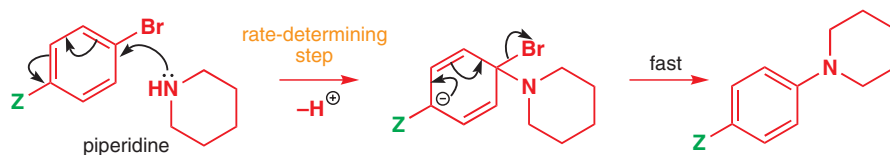
■ Note carefully that this is an *inductive effect*: there are no arrows to be drawn to show how fluorine withdraws electrons—it does it just by polarizing C–F bonds towards itself. Contrast the electron-withdrawing effect of the nitro group, which works (mainly) by conjugation.



Fluoride accelerates the first step through its inductive effect. It is the most electronegative element of all and it stabilizes the anionic intermediate, assisting the acceptance of electrons by the benzene ring.

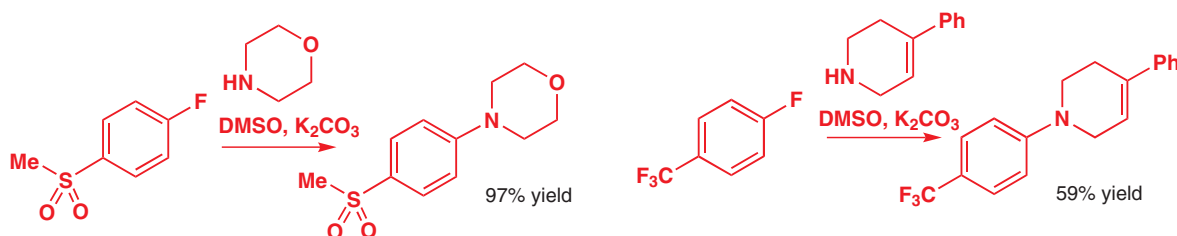
The activating anion-stabilizing substituent

We have used nitro groups very extensively so far because they are the best at stabilizing the anionic intermediate. Others that work include carbonyl, cyanide, and sulfur-based groups such as sulfoxides and sulfones. A direct comparison of the different groups Z that can assist the displacement of bromide (by the secondary amine piperidine in this example) is shown in the margin.



All the compounds react more slowly than the nitro compound. We have already mentioned (Chapters 8 and 21) the great electron-withdrawing power of the nitro group—here is a new measure of that power. The sulfone reacts 18 times slower, the nitrile 32 times slower, and the ketone 80 times slower.

Nitro is the best activating group, but the others will all perform well, especially when combined with a fluoride rather than a bromide as the leaving group. Here are two reactions that work well in a preparative sense with other anion-stabilizing groups. Note that the trifluoromethyl group works by using only its powerful inductive effect.



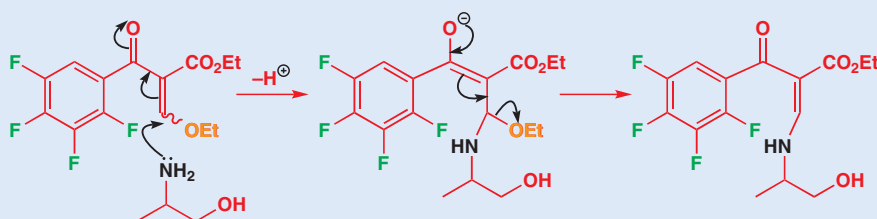
● To summarize

An anion-stabilizing (electron-withdrawing) group *ortho* or *para* to a potential leaving group can be used to facilitate nucleophilic aromatic substitution.

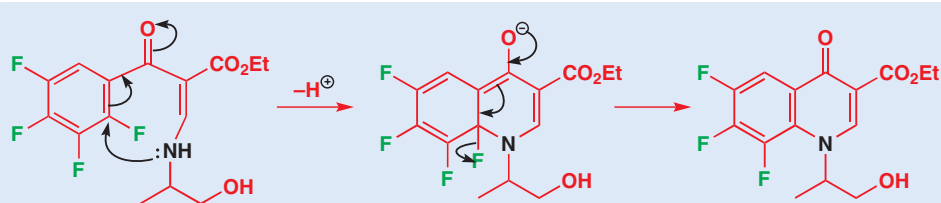
Conjugate and nucleophilic aromatic substitution reactions in action: the synthesis of an antibiotic

We want to convince you that this chemistry is useful and also that it works in more complicated molecules so we are going to describe in part the preparation of the antibiotic ofloxacin. The sequence starts with an aromatic compound having four fluorine atoms. Three are replaced sequentially by nucleophiles and the last is present in the antibiotic itself.

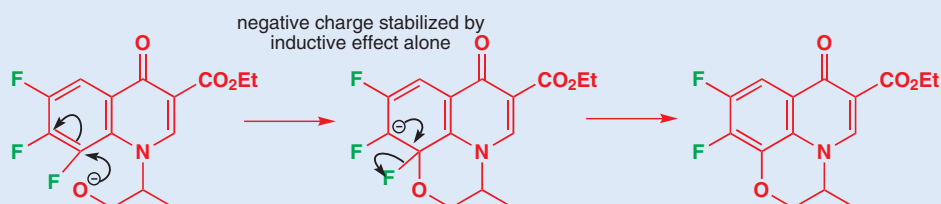
The first reaction is a conjugate substitution of the ethoxy group marked in orange. An amino alcohol is used as the nucleophile and it is the more nucleophilic amino group (rather than the hydroxyl group) that adds to the alkene.



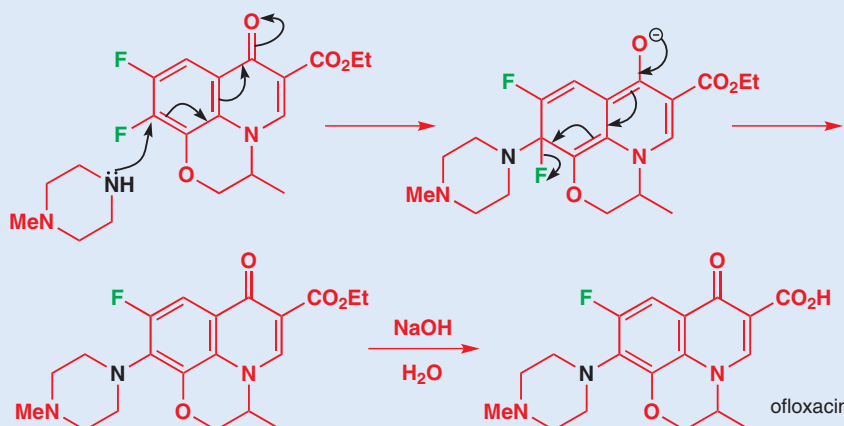
Now for the first nucleophilic aromatic substitution. The amino group attacks in the position *ortho* to the carbonyl group so that an enolate intermediate can be formed. The first fluoride is expelled in the elimination step.



Treatment with base (NaH can be used) now converts the OH group into an alkoxide, which takes part in the next aromatic nucleophilic substitution. In this reaction we are attacking the position *meta* to the ketone so we cannot put the negative charge on the oxygen atom. The combined inductive effect of the remaining three fluorines is enough to stabilize the anion.



Only two fluorines are left, and one of these is now displaced by an external nucleophile—an amine. The site of attack of the amine is determined by the need to stabilize the charge in the intermediate, which is an enolate.

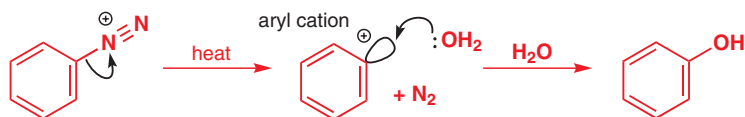


All that is left is to hydrolyse the ester to the free acid with aqueous base (Chapter 10). Every single reaction in this quite complicated sequence is one that you have met earlier in the book, and it illustrates the power of simple organic mechanisms to allow chemists to make important life-saving compounds.

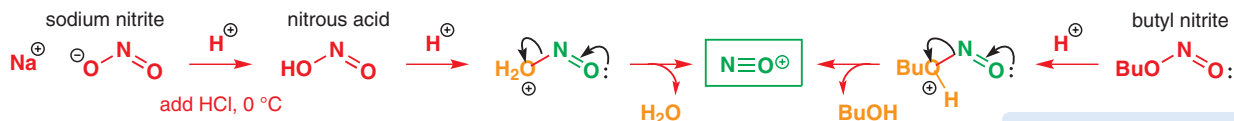
Nucleophilic substitution on aromatic rings is possible by alternative mechanisms as well. We will now turn to these.

The S_N1 mechanism for nucleophilic aromatic substitution: diazonium compounds

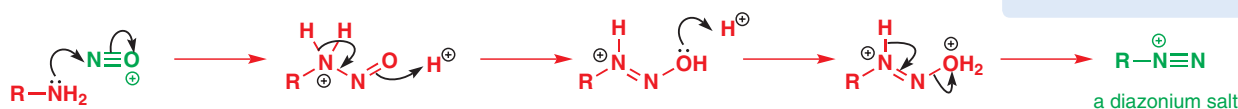
If we really want to make aromatic compounds undergo nucleophilic substitution in a general way, the way to do it is to use absolutely the best leaving group of all—nitrogen gas. In fact, the diazonium compound below is so good at nucleophilic aromatic substitution that it does so even without activating groups. On warming, the nitrogen molecule just departs, leaving behind a cation, which is captured by a nucleophile, in this case water. Do you find this reminiscent of the S_N1 reaction? We hope so.



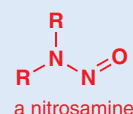
Before we talk about this group of aromatic S_N1 reactions in more detail, let's consider how to make the diazonium salt. The reagent we need is the reactive nitrogen electrophile NO⁺. You met NO⁺ in Chapter 20, but to remind you, it forms when the nitrite anion (usually sodium nitrite) is treated with acid at around 0 °C. Protonation of nitrite gives nitrous acid, HONO; protonation again gives a cation, which can lose water to form NO⁺. Butyl nitrite (or other alkyl nitrites) can also be used as a source of NO⁺.



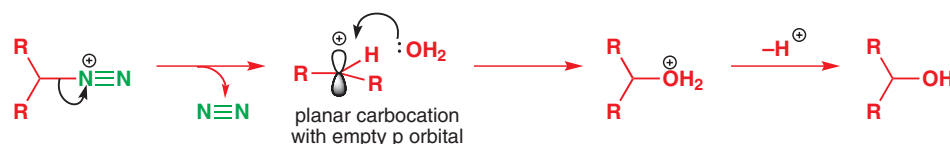
A diazonium salt is formed when NO⁺ reacts with an amine. The lone pair of the amine attacks the NO⁺ cation, and then water is lost. The mechanism is actually quite simple, but it does involve a lot of proton transfers. There is, of course, an anion associated with the nitrogen cation, and this will be the conjugate base (Cl⁻ usually) of the acid used to form NO⁺. This reaction is known as *diazotization*.



If the amine is secondary, water can't be eliminated and a *nitrosamine* forms.



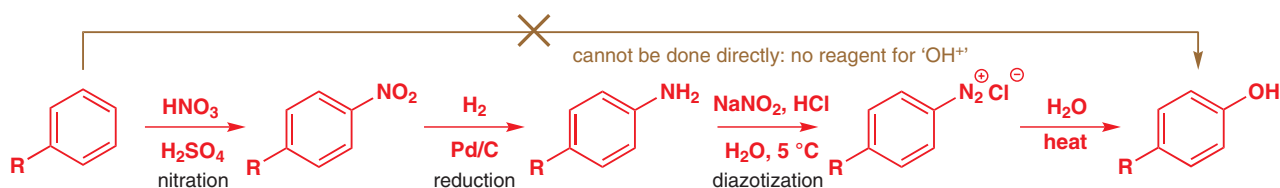
If the amine is an *alkyl* amine, this diazonium salt is very unstable and immediately loses nitrogen gas to give a planar carbocation, which normally reacts with a nucleophile in an S_N1 process (Chapter 15), loses a proton in an E1 process (Chapter 17), or rearranges (Chapter 36). It may, for example, react with water to give an alcohol:



Interactive mechanism for formation of diazonium salt

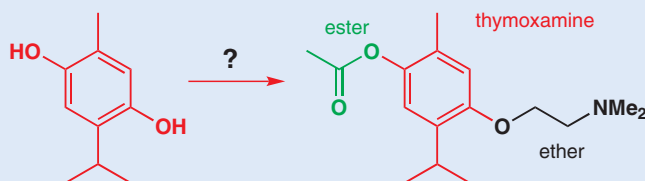
If the amine is an *aryl* amine, then the reaction you saw at the beginning of this section will take place and a phenol will form. This is in fact rather a useful reaction as it is difficult to add an oxygen atom to a benzene ring by normal electrophilic substitution: there is no good reagent for OH⁺. A nitrogen atom can be added easily by nitration, and reduction and diazotization provide a way of replacing the nitro group by a hydroxyl group.

➡ We alluded to this sequence at the end of Chapter 21.

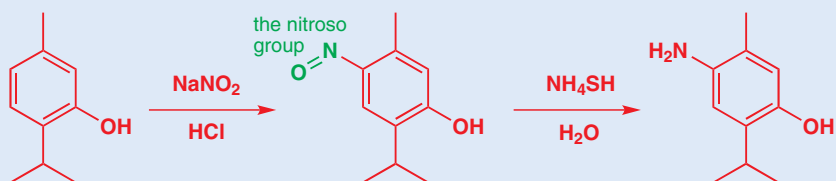


Substitution reactions in the synthesis of a drug

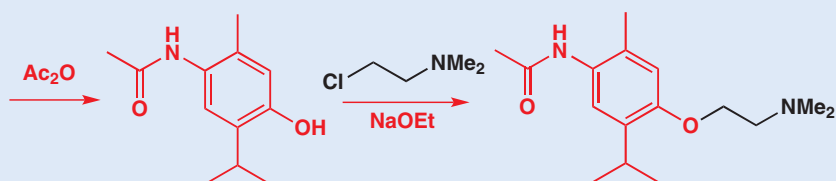
The synthesis of the drug thymoxamine (Moxisylyte) provides a practical example of how this reaction can be used.



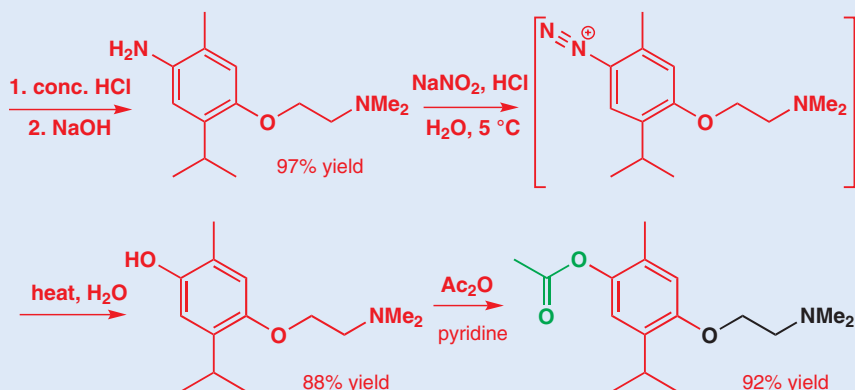
It seems obvious to make this compound by alkylation and acylation of a dihydroxybenzene, but how are we to make sure that the acylation and alkylation go on the right OH groups? French pharmaceutical chemists had an ingenious answer: start with a compound having only one OH group, alkylate that, and only then introduce the second using the diazonium salt method. They used a simple phenol and introduced nitrogen as a nitroso (NO) rather than a nitro (NO_2) group. This means using the same reagent as we have been using for diazotization. These were the first two steps.



The reduction of NO is easier than that of NO_2 , and H_2S is enough to do the job. The amine can now be converted to an amide to lessen its nucleophilicity so that alkylation of the phenol occurs cleanly—a form of protection (see Chapter 23).



Finally, the amide must be hydrolysed, the amino converted into an OH group by diazotization and hydrolysis, and the new phenol acetylated.

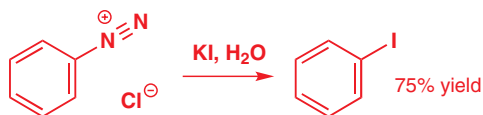


However, an aryl carbocation is much less stable than an alkyl carbocation because its empty orbital is an sp^2 rather than a p orbital. This makes the loss of nitrogen slower. If the diazotization is done at temperatures around 0°C (classically at 5°C), the diazonium salt is stable and can be reacted with various nucleophiles other than water.

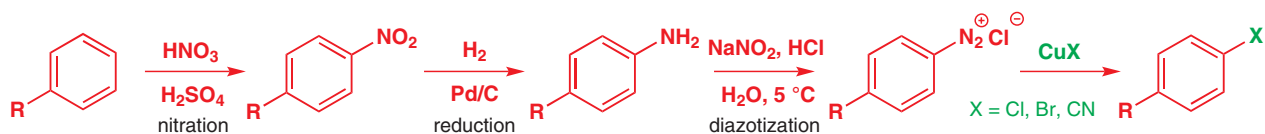
Other nucleophiles

Aryl iodides are not as easy to make by electrophilic substitution as aryl chlorides or bromides because iodine is not reactive enough to attack benzene rings. But adding potassium iodide to the diazonium salt gives an aryl iodide by nucleophilic aromatic substitution.

■ Aryl iodides have wide utility in the coupling chemistry, catalysed by Pd and other transition metals, that you will meet in Chapter 40.



Other nucleophiles, such as chloride, bromide, and cyanide, are best added as copper(I) salts. Since aromatic amines are usually made by reduction of nitro compounds, a common sequence of reactions goes like this:

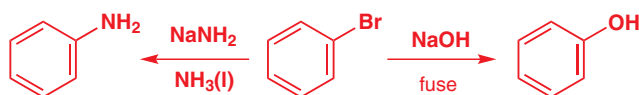


As often in aromatic chemistry, it's the versatility of the nitro group that makes this sequence work—easy introduction by electrophilic substitution, easy reduction, and easy nucleophilic substitution of its diazonium derivative.

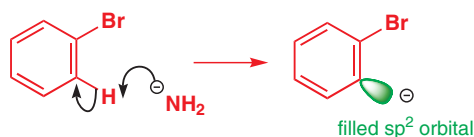
The benzyne mechanism

We now need to introduce you to one last mechanism for aromatic nucleophilic substitution and you may well feel that this is the weirdest mechanism you have yet seen with the most unlikely intermediate ever! For our part, we hope to convince you that this mechanism is not only possible but also useful.

Earlier in this chapter we said that the displacement by nucleophiles of bromide from bromobenzene does not occur. In fact substitution reactions of bromobenzene *can* occur but only under the most vigorous conditions, such as when bromobenzene and NaOH are melted together (fused) at very high temperature. A similar reaction with the very powerful reagent NaNH₂ (which supplies NH₂⁻ ion) also happens, at a rather lower temperature.

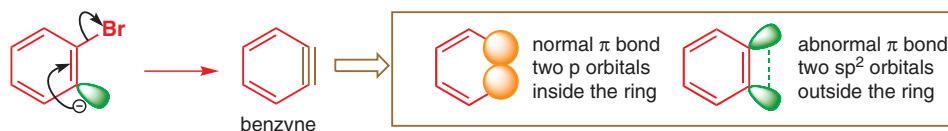


These reactions were known for a long time before anyone saw what was happening. They do not happen by an S_N2 mechanism, as we explained earlier, and they can't happen by the addition–elimination mechanism because there is nothing to stabilize the negative charge in the intermediate. The first clue to the true mechanism is that all the nucleophiles that react in this way are very basic. They start the reaction off by removing a proton *ortho* to the leaving group.



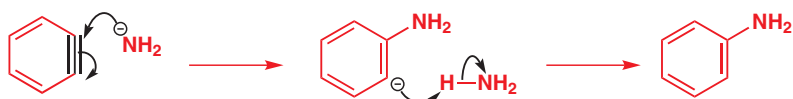
The carbanion is in an sp² orbital in the plane of the ring. Indeed, this intermediate is very similar to the aryl cation intermediate in the S_N1 mechanism from diazonium salts. That had no electrons in the sp² orbital; the carbanion has two. Why should this proton be removed rather than any other? The bromine atom is electronegative and the C–Br bond is in the plane of the sp² orbital and removes electrons from it. The stabilization is nonetheless weak and only exceptionally strong bases will do this reaction.

The next step is the loss of bromide ion in an elimination reaction. This is the step that is difficult to believe as the intermediate we are proposing looks impossible. The orbitals are bad for the elimination too—it is a *syn*- rather than an *anti*-periplanar elimination. But it happens.



The intermediate is called benzyne as it is an alkyne with a triple bond in a benzene ring. But what does this triple bond mean? It certainly isn't a normal alkyne as these are linear. In fact one π bond is normal—it is just part of the aromatic system. One π bond—the new one—is abnormal and is formed by overlap of two sp² orbitals outside the ring. This external π bond

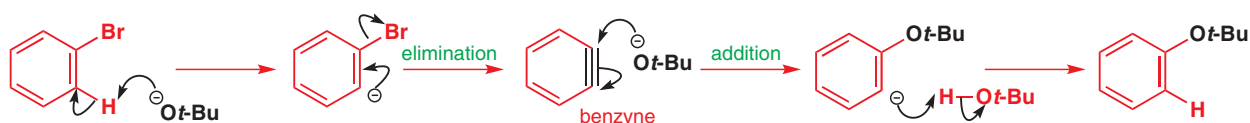
is very weak and benzyne is a very unstable intermediate. Indeed, when the structure was proposed few chemists believed it and some pretty solid evidence was needed before they did. We shall come to that shortly, but let us first finish the mechanism. Unlike normal alkynes, benzyne is electrophilic as the weak third bond can be attacked by nucleophiles.



The whole mechanism from bromobenzene to aniline involves an elimination to give benzyne followed by an addition of the nucleophile to the triple bond of benzyne. In many ways, this mechanism is the reverse of the normal addition–elimination mechanism for nucleophilic aromatic substitution and it is sometimes called the **elimination–addition mechanism**.

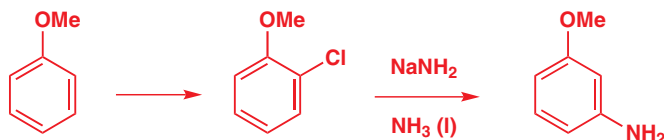
Any nucleophile basic enough to remove the *ortho* proton can carry out this reaction. Known examples include oxyanions, amide anions (R_2N^-), and carbanions. The rather basic alkoxide *t*-butoxide will do the reaction on bromobenzene if the potassium salt is used in the dipolar aprotic solvent DMSO to maximize reactivity.

■ DMSO (see p. 255) solvates K^+ but not RO^- .

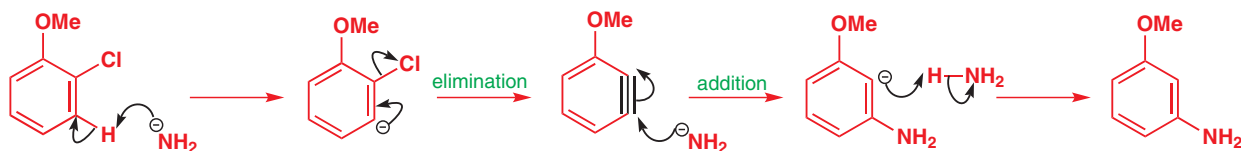


Interactive mechanism for benzyne formation and reaction

One rather special feature of the benzyne mechanism allows us to be certain that this proposed mechanism is correct, and this is the fact that the triple bond could in principle be attacked by nucleophiles at either end. This is of no consequence with bromobenzene as the products would be the same, but we can make the ends of the triple bond different and then we see something interesting. *ortho*-Chloro aryl ethers are easy to prepare by chlorination of the ether (Chapter 21). When these compounds are treated with $NaNH_2$ in liquid ammonia, a single amine is formed in good yield.

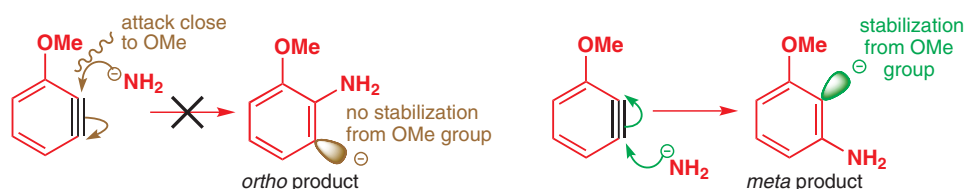


The new amino group finds itself in the *meta* position even though the chlorine was at the *ortho* position. It would be very difficult to explain this other than by the benzyne mechanism. Using the same elimination–addition sequence, this must be the mechanism:



■ Steric hindrance is not nearly as important in electrophilic substitution or in nucleophilic substitution by the addition–elimination mechanism. In both of these reactions, the reagent is attacking the p orbital at *right angles* to the ring and is some distance from an *ortho* substituent.

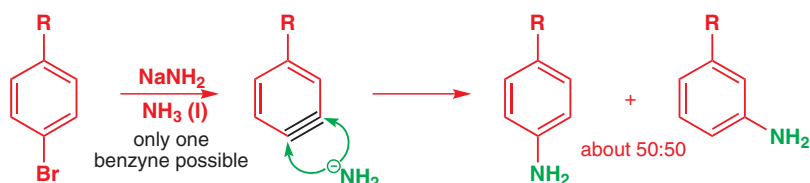
That shows *how* the *meta* product might be formed, but *why* should it be formed? Attack could also occur at the *ortho* position, so why is there no *ortho* product? There are two reasons: electronic and steric. Electronically, the anion next to the electronegative oxygen atom is preferred because oxygen is inductively electron-withdrawing. The same factor facilitates deprotonation next to Cl in the formation of the benzyne. Sterically, it is better for the amide anion to attack away from the OMe group rather than come in alongside it. Nucleophilic attack on a benzyne has to occur in the plane of the benzene ring because that is where the orbitals are. This reaction is therefore very sensitive to steric hindrance as the nucleophile must attack in the plane of the substituent as well.



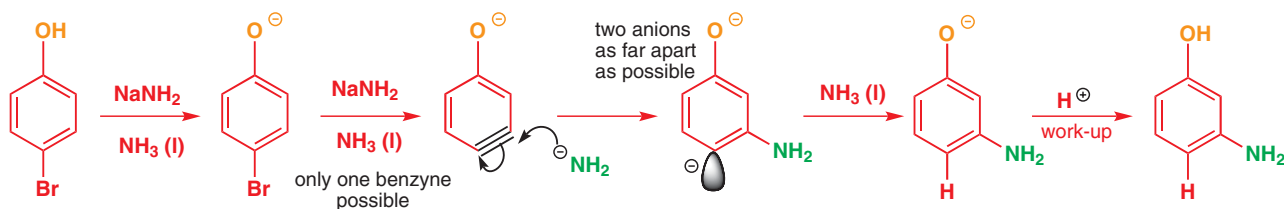
■ Oxygen is an electron-withdrawing group here because the anion is formed in the plane of the ring and has nothing to do with the benzene's π orbitals.

This is a useful way to make amino ethers with a *meta* relationship as both groups are *ortho*, *para*-directing and so the *meta* compounds cannot be made by electrophilic substitution.

para-Disubstituted halides can again give only one benzyne and most of them give mixtures of products. A simple alkyl substituent is too far from the triple bond to have much steric effect.



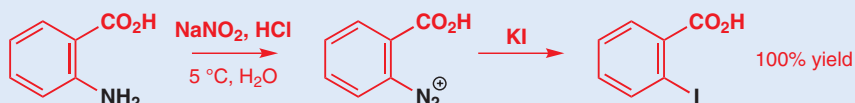
If the substituent is an electron-repelling anion, then the *meta* product is formed exclusively because this puts the product anion as far as possible from the anion already there. This again is useful as it creates a *meta* relationship between two *ortho*, *para*-directing groups.



Other evidence for benzyne as an intermediate

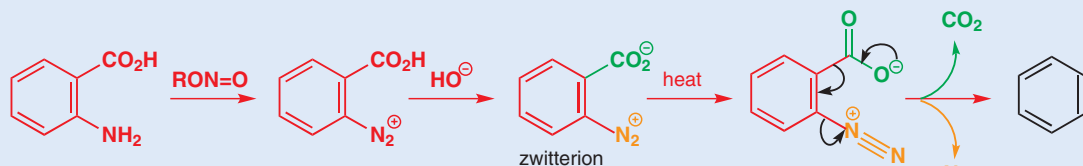
As you would expect, the formation of benzyne is the slow step in the reaction so there is no hope of isolating benzyne from the reaction mixture or even of detect-

ing it spectroscopically. However, it can be made by other reactions where there are no nucleophiles to capture it, for example from this diazotization reaction.



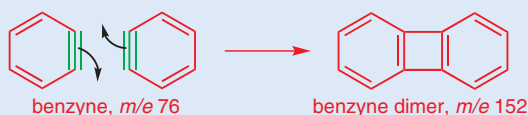
This diazotization is particularly efficient as you can see by the quantitative yield of 2-iodobenzoic acid on capture of the diazonium salt with iodide ion. However, if the same diazonium salt is neutralized with NaOH, it gives a zwitterion with the negative charge on the carboxylate balancing the positive charge

on the diazonium group. This diazotization is done with an alkyl nitrite in an organic solvent to avoid the chance that nucleophiles such as chloride or water might capture the product. When the zwitterion is heated it decomposes in an entropically favourable reaction to give carbon dioxide, nitrogen, and benzyne.



You can't isolate the benzyne because it reacts with itself to give a benzyne dimer having a four-membered ring between two benzene rings. If the zwitterion is injected into a mass spectrometer, there is a peak at 152 for the dimer

but also a strong peak at 76, which is benzyne itself. The lifetime of a particle in the mass spectrometer is about 2×10^{-8} s so benzyne can exist for at least that long in the gas phase.



To conclude...

Alkenes and arenes are usually nucleophiles. This chapter is about the occasions on which they are not, and instead react as electrophiles. Remember that, important though the reactions in this chapter are, the principal reactivity you can expect from these compound classes is nucleophilicity.

The table below summarizes these reactions and also other similar ones you will find elsewhere in the book.

Page	Type of alkene	Example	Reaction
500	unsaturated carbonyl compounds		conjugate addition
510	unsaturated nitriles and nitroalkenes		conjugate addition
511	enones, etc. with β leaving group		conjugate substitution
513	unsaturated carbonyl		nucleophilic epoxidation
515	aryl chlorides/fluorides/ethers with <i>ortho</i> or <i>para</i> electron-withdrawing groups		nucleophilic aromatic substitution: addition–elimination mechanism
520	aryl cations (from diazonium salts)		nucleophilic aromatic substitution: S_N1 mechanism
525	benzyne		nucleophilic aromatic substitution: elimination–addition mechanism
ch. 26	enolates and enolate equivalents as nucleophiles		conjugate addition

Further reading

F. A. Carey and R. J. Sundberg, *Advanced Organic Chemistry A, Structure and Mechanisms*, 5th edn, Springer, 2007, chapter 9 and *B, Reactions and Synthesis*, chapter 11 also has a discussion of nucleophilic aromatic substitution. B. S. Furniss, A. J. Hannaford, P. W. G. Smith, and A. T. Tatchell, *Vogel's Textbook of Practical*

Organic Chemistry, Longman, 5th edn, 1989, 6.6–6.7 gives many practical examples of the nucleophilic aromatic substitution. P. Wyatt and S. Warren, *Organic Synthesis: Strategy and Control*, Wiley, Chichester, 2007, chapter 9.

Check your understanding



To check that you have mastered the concepts presented in this chapter, attempt the problems that are available in the book's Online Resource Centre at <http://www.oxfordtextbooks.co.uk/orc/clayden2e/>